

PIERRE DELIGNE · D030

Values of L-functions and periods of integrals

P. Deligne

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Edition note

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In this article I state a conjecture (1.8, 2.8) relating the values of certain L-functions at certain integral points to periods of integrals.

The L-functions considered are those of motives—a word to which no precise meaning will be attached. These include, in particular, Artin L-functions, L-functions attached to algebraic Hecke characters (= Grössencharakter of type A_0), and those attached to primitive holomorphic modular forms on the Poincaré upper half-plane (= new forms; all the functions L , L_k , attached to the symmetric powers Sym^k of the corresponding l-adic representation are considered).

This article owes its existence to D. Zagier: to his insistence that a conjecture be sought, and to the experimental confirmation he supplied, as soon as it was formulated, for the functions L_3 and L_4 attached to $\Delta = \sum \tau(n)q^n$ (see [18]). It was this confirmation that gave me the confidence needed to verify that the conjecture was compatible with Shimura's results [13] on values of L-functions of algebraic Hecke characters.

0. Motives. The reader is invited not to consult this section except as needs arise. We recall part of the formalism of motives, due to Grothendieck. For proofs, I refer to [8].

0.1. Grothendieck's definition of motives over a field k has the following form. (a) Let $\mathcal{V}(k)$ be the category of smooth projective varieties over k . One constructs an additive category $\mathcal{M}'(k)$, for which the groups $\text{Hom}(M, N)$ are vector spaces over $\mathbb{Q}$, equipped with

α . A tensor product $\otimes$, associative, commutative, and distributive with respect to addition of objects ("associative" and "commutative" are not properties of the functor $\otimes$, but data subject to certain compatibilities—cf. Saavedra [19]);

β . A contravariant functor H^* from $\mathcal{V}(k)$ to $\mathcal{M}'(k)$, bijective on objects and transforming disjoint sums into sums and products into tensor products (the datum of an isomorphism of functors $H^*(X \times Y) = H^*(X) \otimes H^*(Y)$, compatible with associativity and commutativity).

The essential point is to define $\text{Hom}(H^*(X), H^*(Y))$. For the definition, used—

by Grothendieck, of this group as a group of classes of correspondences between X and Y , see 0.6.

(b) Recall (SGA 4 IV 7.5) that an additive category is Karoubian if every projector (= idempotent endomorphism) is defined by a direct-sum decomposition, and that every additive category has a Karoubian envelope, obtained by formally adjoining the images of projectors. The category $\mathcal{M}_{\text{eff}}(k)$ of effective motives over k is the Karoubian envelope of $\mathcal{M}'(k)$.

(c) The Tate motive $\mathbb{Z}(-1)$ is defined as a suitable direct factor $H^2(\mathbb{P}^1)$ of $H^*(\mathbb{P}^1)$. One checks that the symmetry (deduced from the commutativity of $\otimes$), $\mathbb{Z}(-1) \otimes \mathbb{Z}(-1) \rightarrow \mathbb{Z}(-1) \otimes \mathbb{Z}(-1)$, is the identity, and that the functor $M \rightarrow M \otimes \mathbb{Z}(-1)$ is fully faithful.

(d) The category $\mathcal{M}(k)$ of motives over k is obtained from $\mathcal{M}_{\text{eff}}(k)$ by making the functor $M \rightarrow M \otimes \mathbb{Z}(-1)$ invertible.

Let (n) denote the $(-n)$ th iterate of the auto-equivalence $M \rightarrow M \otimes \mathbb{Z}(-1)$. The category $\mathcal{M}(k)$ contains $\mathcal{M}_{\text{eff}}(k)$ as a full subcategory, and every object of $\mathcal{M}(k)$ is of the form $M(n)$, with M in $\mathcal{M}_{\text{eff}}(k)$ and n an integer.

By definition, if F is an additive functor from $\mathcal{M}'(k)$ to a Karoubian category $\mathcal{A}$, it extends to $\mathcal{M}_{\text{eff}}(k)$. If $\mathcal{A}$ is endowed with an auto-equivalence $A \rightarrow A(-1)$, and the extension of F with an isomorphism of functors $F(M(-1)) = F(M)(-1)$, it extends to $\mathcal{M}(k)$.

EXAMPLE 0.1.1. If k' is an extension of k , and F is the functor $H^*(X) \rightarrow H^*(X \otimes_k k')$ from $\mathcal{M}'(k)$ to $\mathcal{M}'(k')$, one obtains the scalar-extension functor from $\mathcal{M}(k)$ to $\mathcal{M}(k')$. If k' is a finite extension of k , one has Grothendieck's restriction-of-scalars functor

$$\Pi_{k'/k}: \mathcal{V}(k') \rightarrow \mathcal{V}(k): (X \rightarrow \text{Spec}(k')) \mapsto (X \rightarrow \text{Spec}(k') \rightarrow \text{Spec}(k)).$$

It extends by $F: H^*(X) \mapsto H^*(\Pi_{k'/k} X)$, yielding a restriction-of-scalars functor $R_{k'/k}: \mathcal{M}(k') \rightarrow \mathcal{M}(k)$.

EXAMPLE 0.1.2. Let $\mathcal{H}$ be a “cohomology theory” with values in a Karoubian category $\mathcal{A}$, functorial for morphisms in $\mathcal{M}'(k)$. The functor $\mathcal{H}$ extends to $\mathcal{M}_{\text{eff}}(k)$. If $\mathcal{A}$ has a tensor product, $\mathcal{H}$ satisfies a Künneth formula, and tensoring with $\mathcal{H}(\mathbb{Z}(-1))$ is an auto-equivalence of $\mathcal{A}$, then it extends to $\mathcal{M}(k)$. This extension is the functor “realization of a motive in the theory $\mathcal{H}$ ”.

We shall write (n) for the auto-equivalence of $\mathcal{A}$ obtained as the $(-n)$ th iterate of tensoring with $\mathcal{H}(\mathbb{Z}(-1))$. For the determination of $\mathcal{H}(\mathbb{Z}(-1))$ in various theories $\mathcal{H}$, see 3.1.

0.2. We shall use the following realizations: 0.2.1. Betti realization H_B . Corresponding to $k = \mathbb{C}$, $\mathcal{A}$ = vector spaces over $\mathbb{Q}$, $\mathcal{H}$ = rational cohomology: $X \mapsto H^*(X(\mathbb{C}), \mathbb{Q})$; 0.2.2. de Rham realization H_{DR} . Corresponding to k of characteristic 0, $\mathcal{A}$ = vector spaces over k , $\mathcal{H}$ = de Rham cohomology: $X \mapsto H^*(X, \Omega_X^*)$; 0.2.3. l -adic realization H_l . Corresponding to algebraically closed k of characteristic $\neq l$, $\mathcal{A}$ = vector spaces over $\mathbb{Q}_l$, $\mathcal{H}$ = l -adic cohomology: $X \mapsto H^*(X, \mathbb{Q}_l)$.

And their variants: 0.2.4. Hodge realization. Here k is an algebraic closure of $\mathbb{R}$, and $\mathcal{A}$ is the category of vector spaces over $\mathbb{Q}$ whose scalar extension $V \otimes k$ is endowed with a bigrading

$V \otimes k = \bigoplus_{p,q} V^{p,q}$, with $V^{q,p}$ the complex conjugate of $V^{p,q}$. For the cohomology theory, one takes the functor $X \mapsto H^*(X(k), \mathbb{Q})$, endowed with the bigrading of its scalar extension $H^*(X(k), k)$ supplied by Hodge theory. The Betti realization underlies the Hodge realization.

0.2.5. For arbitrary k and a complex embedding σ of k , write $H_\sigma(M)$ for the Betti realization of the motive over $\mathbb{C}$ obtained from M by scalar extension through σ : $k \rightarrow \mathbb{C}$. Let c denote complex conjugation. Transport of structure gives an isomorphism F_∞ : $H_\sigma(M) \cong H_{c\sigma}(M)$, and $F_\infty \otimes c$ sends $H_\sigma^{p,q}$ to $H_{c\sigma}^{q,p}$. If σ is real, F_∞ is an involution of $H_\sigma(M)$, whose complexification interchanges $H_\sigma^{p,q}(M)$ and $H_\sigma^{q,p}(M)$.

The σ -Hodge realization is $H_\sigma(M)$, endowed with its Hodge bigrading and, when σ is real, the involution F_∞ . For $k = \mathbb{Q}$, or $k = \mathbb{R}$ with σ the identity embedding, the index σ is replaced by B . For $k = \mathbb{R}$ and $M = H^*(X)$, the involution F_∞ on $H_B^*(M) = H^*(X(\mathbb{C}), \mathbb{Q})$ is induced by complex conjugation F_∞ : $X(\mathbb{C}) \rightarrow X(\mathbb{C})$.

0.2.6. de Rham realization. de Rham cohomology carries a natural filtration, the Hodge filtration, arising as the abutment of the hypercohomology spectral sequence

$$E_1^{p,q} = H^q(X, \Omega^p) \Rightarrow H_{\text{DR}}^{p+q}(X).$$

This gives a filtration F on $H_{\text{DR}}(M)$.

0.2.7. l-adic realization. If X is a variety over a field k , with algebraic closure $\bar{k}$, the Galois group $\text{Gal}(\bar{k}/k)$ acts by transport of structure on $H^*(X_{\bar{k}}, \mathbb{Q}_l)$. If M is a motive over k and $H_l(M)$ is defined as the l-adic realization of the motive over $\bar{k}$ obtained from it by scalar extension, this gives an action of $\text{Gal}(\bar{k}/k)$ on $H_l(M)$.

0.2.8. Finite adelic realization. For algebraically closed k of characteristic 0, the l-adic cohomologies can be assembled into a finite adelic cohomology

$$H^*(X, A^f) = \left(\prod_l H^*(X, \mathbb{Z}_l) \right) \otimes_{\mathbb{Z}} \mathbb{Q}.$$

The variants 0.2.7 and 0.2.8 may be combined.

0.2.9. In Examples 0.2.1, 0.2.2, 0.2.3 and their variants, the category $\mathcal{A}$ may be replaced by the category $\mathcal{A}^*$ of graded objects of $\mathcal{A}$, using the natural grading of H^* . In order that the Künneth formula yield an isomorphism of functors $\mathcal{H}(M \otimes N) \cong \mathcal{H}(M) \otimes \mathcal{H}(N)$, compatible with associativity and commutativity, the commutativity datum in $\mathcal{A}^*$ must be the one given by the Koszul rule.

0.3. For $k = \mathbb{C}$, the comparison theorem between classical and étale cohomology gives an isomorphism $H^*(X(\mathbb{C}), \mathbb{Q}) \otimes \mathbb{Q}_l \cong H^*(X, \mathbb{Q}_l)$. If X is defined over $\mathbb{R}$, this isomorphism carries F_∞ (0.2.5) to the action (0.2.7) of complex conjugation. For a motive one likewise has $H_B(M) \otimes \mathbb{Q}_l \cong H_l(M)$.

0.4. For $k = \mathbb{C}$, the holomorphic de Rham complex on X^{an} is a resolution of the constant sheaf $\mathbb{C}$. By GAGA one therefore has an isomorphism

$$H^*(X(\mathbb{C}), \mathbb{Q}) \otimes \mathbb{C} = H^*(X(\mathbb{C}), \mathbb{C}) \cong H^*(X^{\text{an}}, \Omega^{*an}) \cong H^*(X, \Omega^*).$$

For a motive one obtains an isomorphism, compatible with the Hodge filtrations in 0.2.4— $F^p = \bigoplus_{p' \geq p} V^{p',q'}$ —and 0.2.6: $H_B(M) \otimes \mathbb{C} \cong H_{\text{DR}}(M)$.

More generally, let M be a motive over a field k and σ a complex embedding of k . Applying the preceding construction to the motive over $\mathbb{C}$ obtained from M by scalar extension gives an isomorphism

$$(0.4.1) \quad H_{\sigma}(M) \otimes \mathbb{C} \cong H_{\text{DR}}(M) \otimes_{k,\sigma} \mathbb{C}.$$

In the particular case $k = \mathbb{Q}$, this gives two natural $\mathbb{Q}$ -structures on the realization of M in complex cohomology: one, $H_B(M)$, linked to the description of cohomology in terms of cycles; the other, $H_{\text{DR}}(M)$, linked to its description in terms of algebraic differential forms.

0.5. The behavior of these realizations and compatibilities under extension of the scalar field is clear. For restriction of scalars, the results are as follows. Let k' be a finite extension of k , let M' be a motive over k' , and let $M = R_{k'/k}(M')$.

$$(0.5.1) \quad H_{\sigma}(M) = \bigoplus_{\tau} H_{\tau}(M'),$$

where the sum extends over the set $J(\sigma)$ of complex embeddings of k' extending σ . This isomorphism is compatible with the Hodge bigradings and with F_{∞} .

$$(0.5.2) \quad H_{\text{DR}}(M) = H_{\text{DR}}(M') \text{ (restriction of scalars from } k' \text{ to } k).$$

This isomorphism is compatible with the Hodge filtration.

$$(0.5.3) \quad H_l(M) = \text{Ind } H_l(M') \text{ (the representation induced from } \text{Gal}(\bar{k}/k') \text{ to } \text{Gal}(\bar{k}/k)).$$

(0.5.4) Via the isomorphism $k' \otimes_{k,\sigma} \mathbb{C} = \mathbb{C}^{J(\sigma)}$, and the resulting isomorphism $H_{\text{DR}}(M') \otimes_{k,\sigma} \mathbb{C} = \bigoplus_{\tau \in J(\sigma)} H_{\text{DR}}(M') \otimes_{k',\tau} \mathbb{C}$, (0.4.1) for M and σ is the sum of the isomorphisms (0.4.1) for M' and τ ($\tau \in J(\sigma)$):

$$\begin{array}{ccc} H_{\sigma}(M) \otimes \mathbb{C} & \xrightarrow{\cong} & H_{\text{DR}}(M) \otimes_{\{k,\sigma\}} \mathbb{C} \\ \parallel (0.5.1) & & \parallel (0.5.2) \\ \bigoplus_{\tau} H_{\tau}(M') \otimes \mathbb{C} & \xrightarrow{\cong} & \bigoplus_{\tau} H_{\text{DR}}(M') \otimes_{\{k',\tau\}} \mathbb{C}. \end{array}$$

0.6. For X smooth and projective over k , let $Z^d(X)$ be the $\mathbb{Q}$ -vector space with basis the irreducible closed subschemes of codimension d in X , and let $Z_R^d(X)$ be its quotient by an equivalence relation R . In characteristic 0, one of Grothendieck's definitions of motives is obtained by setting, for connected X and Y ,

$$\text{Hom}(H^*(Y), H^*(X)) = Z_R^{\dim(Y)}(X \times Y),$$

with R equal to homological equivalence. In characteristic p there are two difficulties to note: it is not known whether homological equivalence in l -adic cohomology ($l \neq p$) is independent of l , and defining the class of a cycle raises problems in crystalline cohomology.

0.7. Let X be a smooth projective complex variety. A Hodge cycle of codimension d on X will mean an element of $H^{2d}(X(\mathbb{C}), \mathbb{Q})$ of type (d, d) , or equivalently an element of $H^{2d}(X(\mathbb{C}), \mathbb{Q})(d)$ of type $(0, 0)$.

Let k be an algebraically closed field and X a smooth projective variety over k . Set

$$H^{2d}(X, k \times A^f)(d) = H_{DR}^{2d}(X)(d) \times H^{2d}(X, A^f)(d).$$

For $k = \mathbb{C}$, the image in

$$H^{2d}(X, k \times A^f)(d) = H^{2d}(X(\mathbb{C}), \mathbb{Q})(d) \otimes (k \times A^f)$$

of a Hodge cycle is again called a Hodge cycle. For algebraically closed k admitting complex embeddings, an absolute Hodge cycle of codimension d on X is an element of $H^{2d}(X, k \times A^f)(d)$ whose image in $H^{2d}(X \otimes_k \mathbb{C}, \mathbb{C} \times A^f)(d)$ is a Hodge cycle for every complex embedding σ of k . One verifies the following.

PROPOSITION 0.8. (i) The $\mathbb{Q}$ -vector space $Z_{ha}^d(X)$ of absolute Hodge cycles is invariant under extension of scalars from k to an algebraically closed field k' that still admits a complex embedding.

(ii) Suppose k is algebraically closed of characteristic 0 and X is defined over an algebraically closed subfield k_0 of k admitting a complex embedding, with $X = X_0 \otimes_{k_0} k$. Set

$$Z_{ha}^d(X) = Z_{ha}^d(X_0) \subset H^{2d}(X_0, k_0 \times A^f)(d) \subset H^{2d}(X, k \times A^f)(d).$$

By (i), this definition is independent of the choices of X_0 and k_0 .

(iii) If X is defined over a subfield k_0 of k , with $X = X_0 \otimes_{k_0} k$, then $\text{Aut}(k/k_0)$, acting on $H^{2d}(X, k \times A^f)(d)$, stabilizes $Z_{ha}^d(X)$. Its action on $Z_{ha}^d(X)$ factors through a finite group corresponding to a finite extension k'_0 of k_0 . Set $Z_{ha}^d(X_0) = Z_{ha}^d(X)^{\text{Aut}(k/k'_0)}$.

0.9. A useful notion of motive is obtained by setting, for connected X and Y ,

$$\text{Hom}(H^*(Y), H^*(X)) = Z_{ha}^{\dim(Y)}(X \times Y).$$

The Künneth components of the diagonal of $X \times X$ are absolute Hodge cycles. This makes it possible to decompose $H^*(X)$ into a sum of motives $H^i(X)$, to grade the category of motives—with $H^i(X)$ of weight i —and to modify the commutativity constraint for $\otimes$ as in [19, VI, 4.2.1.4]. With these conventions, the tensor category of motives over k is Tannakian and isomorphic to the category of representations of a reductive proalgebraic group.

For $k = \mathbb{C}$, the following conjecture, weaker than the Hodge conjecture, is equivalent to saying that the Hodge-realization functor is an equivalence between the category of motives in 0.9 and the category of Hodge structures that occur as direct factors of the cohomology of algebraic varieties, or are obtained from such factors by Tate twist.

Hope 0.10. Every Hodge cycle is absolute.

If X is an abelian variety with complex multiplication by an imaginary quadratic field K , with $\text{Lie}(X)$ free over $k \otimes K$, the methods of B. Gross [7] prove that certain nontrivial Hodge cycles are absolute Hodge cycles.

Added in proof: starting from this special case, I have been able to verify 0.10 for abelian varieties. I have also verified that the tensor category of motives (0.9) generated by $H^n(X)$, for X an abelian variety, contains the motives $H^*(X)$ for X a K3 surface or a Fermat hypersurface.

0.11. The principal defect of definition 0.9 of motives is that it does not lend itself to reduction modulo p . It is not known whether a motive in the sense of 0.9 over a number field F gives a compatible system of l -adic representations of $\text{Gal}(\bar{F}/F)$.

0.12. We shall use the word “motive” freely, without concerning ourselves with placing the motives under consideration within Grothendieck’s framework. What matters for us is to have realizations $\mathcal{H}(M)$, for the theories $\mathcal{H}$ considered in 0.2, and to have for these groups the same formalism as for the $\mathcal{H}^*(X)$.

1. Statement of the conjecture (rational case).

1.1. Let M be a motive over $\mathbb{Q}$. We shall assume that the l -adic realizations $H_l(M)$ of M form a strictly compatible system of l -adic representations, in the sense of Serre [11, I.11]. Namely, there is a finite set S of prime numbers such that each $H_l(M)$ is unramified outside $S \cup \{l\}$, and, if $F_p \in \text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ denotes a geometric Frobenius element at p —the inverse of a Frobenius substitution φ_p —then the polynomial $\det(1 - F_p t, H_l(M)) \in \mathbb{Q}_l[t]$, for $p \notin S \cup \{l\}$, has rational coefficients and is independent of l . Let $Z_p(M, t)$ denote its inverse, and set $L_p(M, s) = Z_p(M, p^{-s})$.

The Dirichlet series with rational coefficients given by the Euler product

$$L_S(M, s) = \prod_{p \notin S} L_p(M, s)$$

converges for $\text{Re}(s)$ sufficiently large. For arbitrary s , $L_S(M, s)$ is defined by analytic continuation, which is expected to exist.

Our aim is to state a conjecture giving the value of $L_S(M, s)$ at certain integral points, up to a rational factor. Since p^{-s} is rational for integral s , the choice of S is immaterial, except that enlarging S can introduce unwanted zeros.

1.2. To write the conjectural functional equation of L -functions properly, the Euler product $L_S(M, s)$ must be completed by local factors at $p \in S$ and at infinity. Defining the local factors at $p \in S$ requires an additional hypothesis, which we assume:

(1.2.1) Let p be a prime number, $D_p \subset \text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ a decomposition group at p , $I_p \subset D_p$ its inertia subgroup, and $F_p \in D_p$ a geometric Frobenius. The polynomial $\det(1 - F_p t, H_l(M)^{I_p}) \in \mathbb{Q}_l[t]$, for $l \neq p$, has rational coefficients and is independent of l .

Set $Z_p(M, t) = \det(1 - F_p t, H_l(M)^{I_p})^{-1} \in \mathbb{Q}(t)$, and $L_p(M, s) = Z_p(M, p^{-s})$. Define

$$(1.2.2) \quad L(M, s) = \prod_p L_p(M, s).$$

The factor at infinity $L_\infty(M, s)$, essentially a product of Γ -functions, depends on the Hodge realization of M —in fact only on the isomorphism class of the complex vector space $H_B(M) \otimes \mathbb{C}$, endowed with its Hodge decomposition and the involution F_∞ . Its definition is recalled in 5.2.

Setting $\Lambda(M, s) = L_\infty(M, s)L(M, s)$, the conjectural functional equation of L -functions is

$$(1.2.3) \quad \Lambda(M, s) = \varepsilon(M, s)\Lambda(\check{M}, 1 - s),$$

where $\check{M}$ is the dual of M —its realizations are dual to those of M —and $\varepsilon(M, s)$, as a function of s , is the product of a constant and an exponential. Its definition is recalled in 5.2. It depends on an additional hypothesis.

DEFINITION 1.3. An integer n is critical for M if neither $L_\infty(M, s)$ nor $L_\infty(\check{M}, 1 - s)$ has a pole at $s = n$.

Our aim is to conjecture the value of $L(M, n)$ for critical n , up to multiplication—

by a rational number. We have $L(M(n), s) = L(M, n + s)$ (3.1.2), and likewise for L_∞ . This allows us to consider only the numbers $L(M) =_{\text{def}} L(M, 0)$. We shall say that M is critical if 0 is critical for M . One verifies that M is critical if and only if the Hodge numbers $h^{p,q} =_{\text{def}} \dim H^{p,q}(M)$ of M , for $p \neq q$, can be nonzero only for (p, q) in the shaded part of the diagram below, and F_∞ acts on $H^{p,p}$ as the identity if $p < 0$ and as -1 if $p \geq 0$.

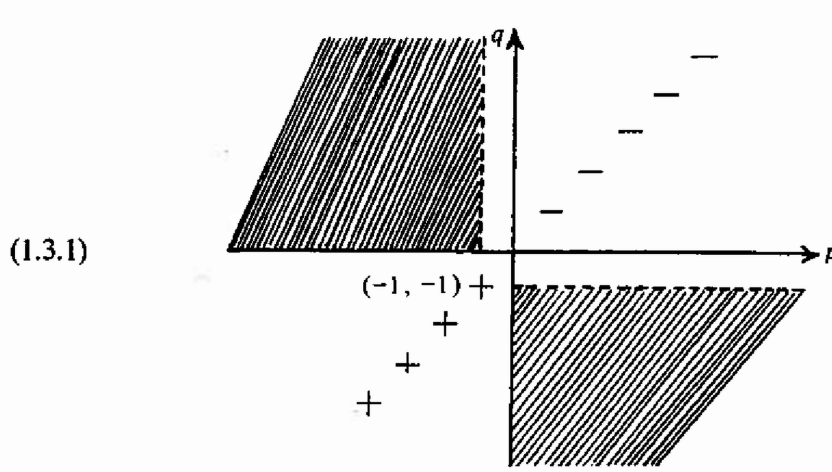


Diagram (1.3.1), certified presentation derivative; the lossless authority crop is retained with the sources.

Suppose that M is homogeneous of weight w , so $h^{p,q} = 0$ for $p + q \neq w$, and set $\Re(M) = -w/2$. The Weil conjecture implies that, for S sufficiently large, the Dirichlet series $L_S(M, s)$ converges absolutely when $\Re(M) + \Re(s) > 1$. When $\Re(M) + \Re(s) = 1$, one is on the boundary of the half-plane of convergence, and one conjectures (a) that $L_S(M, s)$ does not vanish; (b) that $L_S(M, s)$ is holomorphic, except when M has even weight $-2n$ and contains $\mathbb{Z}(n)$ as a factor, in which case one expects a pole at $s = 1 - n$, a noncritical value.

The analogy with function fields and the known cases suggest that the local factors $L_p(M, s)$, for arbitrary p including infinity, have poles only when $\Re(M) + \Re(s) \leq 0$. If this is so, (a), (b), and the conjectural functional equation (1.2.3) imply that $L(M)$ is neither zero nor infinite for critical M with $\Re(M) \neq 1/2$. When $\Re(M) = 1/2$, $L(M)$ sometimes vanishes, and Conjecture 1.8 is then vacuous.

PROPOSITION 1.4. Let M be a motive over $\mathbb{R}$. Under the isomorphism (0.4.1)

$$H_B(M) \otimes \mathbb{C} \xrightarrow{\sim} H_{DR}(M) \otimes_{\mathbb{R}} \mathbb{C},$$

$H_{DR}(M)$ identifies with the subspace of $H_B(M) \otimes \mathbb{C}$ fixed by $c \mapsto F_\infty \bar{c}$.

Take $M = H^*(X)$. Complex conjugation on $H_{DR}(M)_{\mathbb{C}} = H^*(X_{\mathbb{C}}, \Omega^*)$ follows functorially from the antilinear automorphism F_∞ of the scheme $X_{\mathbb{C}}$. Tracing back through the composite arrow (0.4) defining I , it is identified with the involution induced by the automorphism (F_∞, F_∞^*) of $(X(\mathbb{C}), \Omega^{*an})$, and then with the composite of $F_\infty^*: H^*(X(\mathbb{C}), \mathbb{C}) \rightarrow H^*(X(\mathbb{C}), \mathbb{C})$ and complex conjugation on the coefficients. This proves 1.4.

1.5. For a motive M over $\mathbb{R}$, let $H_B^+(M)$, respectively $H_B^-(M)$, denote the subspace of $H_B(M)$ fixed by F_∞ , respectively the subspace on which $F_\infty = -1$. Set $d(M) = \dim H_B(M)$ and $d^\pm(M) = \dim H_B^\pm(M)$. If M is a motive over k and σ is a complex embedding of k factoring through $\mathbb{R}$, write $H_\sigma^\pm(M)$, $d_\sigma^\pm(M)$, and $d_\sigma(M)$ for these objects attached to the motive over $\mathbb{R}$ obtained from M through σ . For $k = \mathbb{Q}$, the reference to σ is omitted.

The following corollary follows immediately from 1.4 and from the fact that both F_∞ and complex conjugation interchange $H^{p,q}$ and $H^{q,p}$.

COROLLARY 1.6. Let M be a motive over $\mathbb{R}$. For the real structure $H_{\text{DR}}(M)$ on $H_B(M) \otimes \mathbb{C}$, the subspaces $H^{p,q}$ are defined over $\mathbb{R}$; the subspace $H_B^+(M) \otimes \mathbb{R}$ is real, and $H_B^-(M) \otimes \mathbb{R}$ is purely imaginary.

1.7. In the remainder of this section we consider only motives over $\mathbb{Q}$ and, unless explicitly stated otherwise, assume them homogeneous. If their weight w is even, we also assume that F_∞ acts on $H^{\text{pp}}(M)$, where $w = 2p$, by a scalar, either $+1$ or -1 . This hypothesis holds for critical M . Since F_∞ interchanges $H^{p,q}$ and $H^{q,p}$, the dimensions $d^+(M)$ and $d^-(M)$ are, in some order, $\sum_{p>q} h^{p,q}$ and $\sum_{p\geq q} h^{p,q}$. In particular, these dimensions equal those of the subspaces F^+ and F^- occurring in the Hodge filtration. Setting $H_{\overline{\text{DR}}}^\pm(M) = H_{\text{DR}}(M)/F^\mp$, one also has $\dim H_{\overline{\text{DR}}}^\pm(M) = d^\pm(M)$.

It follows that the composite maps

$$(1.7.1) \quad I^\pm: H_B^\pm(M)_\mathbb{C} \rightarrow H_B(M)_\mathbb{C} \xrightarrow{-I, \cong} H_{\text{DR}}(M)_\mathbb{C} \rightarrow H_{\overline{\text{DR}}}^\pm(M)_\mathbb{C}$$

are isomorphisms. Set

$$(1.7.2) \quad c^\pm(M) = \det(I^\pm), \quad (1.7.3) \quad \delta(M) = \det(I),$$

where the determinant is computed in rational bases of $H_B^\pm$ and $H_{\overline{\text{DR}}}^\pm$, respectively H_B and H_{DR} . The definition of $\delta(M)$ does not require the hypotheses imposed on M .

By 1.6, I^+ is real—that is, it induces $I^+: H_B^+(M)_\mathbb{R} \rightarrow H_{\overline{\text{DR}}}^+(M)_\mathbb{R}$ —whereas I^- is purely imaginary. Thus $c^+(M)$, $i^{d^-(M)}c^-(M)$, and $i^{d^-(M)}\delta(M)$ are nonzero real numbers. Up to multiplication by a rational number, they depend only on M .

The periods of M are classically the numbers $\langle \omega, c \rangle$ with $\omega \in H_{\text{DR}}(M)$ and $c \in H_B(M)^\vee$. For example, if X is an algebraic variety over $\mathbb{Q}$, ω is an n -form on X defined over $\mathbb{Q}$, and c is an n -cycle on $X(\mathbb{C})$, then $\langle \omega, c \rangle = \int_c \omega$ is a period of $H^n(X)$. Let us express $c^\pm(M)$ in terms of periods. The dual of $H_{\overline{\text{DR}}}^\pm(M)$ is the subspace $F^\pm$ of $H_{\text{DR}}(M^\vee)$, where $M^\vee$ is the dual motive. If the chosen basis of $H_{\overline{\text{DR}}}^\pm(M)$ has dual basis (ω_i) in $F^\pm(H_{\text{DR}}(M^\vee))$, and (c_j) is the chosen basis of $H_B^\pm(M)$, then the matrix of $I^\pm$ is $(\langle \omega_i, c_j \rangle)$, and $c^\pm(M) = \det(\langle \omega_i, c_j \rangle)$.

Conjecture 1.8. If M is critical, $L(M)$ is a rational multiple of $c^+(M)$.

2. Statement of the conjecture (general case).

2.1. Until now we have considered only L -functions given by Dirichlet series with rational coefficients. To do better, we must consider motives with coefficients in number fields.

There are two equivalent ways to construct the category of motives over k ,

with coefficients in a number field E , from the category of motives over k . The construction applies to any additive Karoubian category in which $\text{Hom}(X, Y)$ is a $\mathbb{Q}$ -vector space.

A. A motive over k with coefficients in E is a motive M over k endowed with an E -module structure $E \rightarrow \text{End}(M)$.

B. The category $\mathcal{M}_{k,E}$ of motives over k with coefficients in E is the Karoubian envelope of the category whose objects are motives over k —when regarded as an object of $\mathcal{M}_{k,E}$, a motive M is denoted M_E —and whose morphisms are

$$\text{Hom}(X_E, Y_E) = \text{Hom}(X, Y) \otimes E.$$

Passage from B to A. For a motive X and a finite-dimensional $\mathbb{Q}$ -vector space V , let $X \otimes V$ denote the motive isomorphic to a sum of $\dim(V)$ copies of X and characterized by $\text{Hom}(Y, X \otimes V) = \text{Hom}(Y, X) \otimes V$, or equivalently by $\text{Hom}(X \otimes V, Y) = \text{Hom}(V, \text{Hom}(X, Y))$. Associate to M_E the motive $M \otimes E$ with its natural E -module structure.

Passage from A to B. If M has an E -module structure, M_E carries two E -module structures: the one induced by the structure on M and the one carried by every object of $\mathcal{M}_{k,E}$. The object of $\mathcal{M}_{k,E}$ corresponding to M is the largest direct factor of M_E on which these structures agree. More explicitly, $E \otimes E$ is a product of fields, including a copy of E in which $x \otimes 1$ and $1 \otimes x$ both project to x . The corresponding idempotent e acts on M_E , which is an $E \otimes E$ -module, and its image is the object of $\mathcal{M}_{k,E}$ corresponding to M .

Motives with coefficients in E are most often given in form A. Form B has the advantage of remaining meaningful when E is not of finite rank over $\mathbb{Q}$. It is useful for understanding the tensor formalism: tensor products and duals of motives with coefficients in E are defined by functoriality and the formulas

$$X_E \otimes_E Y_E = (X \otimes Y)_E \text{ and } (X_E)^\vee = (X^\vee)_E.$$

In language A, $X \otimes_E Y$ is the largest direct factor of $X \otimes Y$ on which the two E -module structures agree, and $X^\vee$ is the ordinary dual of X endowed with the transposed E -module structure. Applying these observations to vector spaces over $\mathbb{Q}$ rather than to motives recovers the isomorphism between the E -dual of an E -vector space and its $\mathbb{Q}$ -dual, given by

$$\omega \mapsto \text{the form } \text{Tr}_{E/\mathbb{Q}}(\langle \omega, v \rangle).$$

In 0.1.1 we defined restriction and extension functors for the base field k . They carry motives with coefficients in E to motives with coefficients in E . There are also restriction and extension functors for the coefficient field; let F be a finite extension of E .

Extension of coefficients. In language A this is $X \mapsto X \otimes_E F$; in language B it is $X_E \mapsto X_F$.

Restriction of coefficients. In language A one restricts the F -module structure to E .

Care must be taken not to confuse the roles of k and E . A standard example is H_1 of an abelian variety over k with complex multiplication by an order in E . In terms of the abelian variety, taken up to isogeny, the preceding functors become extension of the base field, Weil restriction of scalars, the construction $\otimes_E F$, which multiplies the dimension by $[F:E]$, and restriction to E of the F -module structure.

2.2. Let M be a motive over $\mathbb{Q}$ with coefficients in a number field E . For each prime number l , the l -adic realization $H_l(M)$ is a module over the l -adic completion E_l of E . This completion is the product of the completions E_λ over the prime ideals λ above l , giving a decomposition of $H_l(M)$ into a product of E_λ -modules $H_\lambda(M)$.

One conjectures for the $H_\lambda(M)$ a compatibility analogous to 1.2.1. If it holds, then for each complex embedding σ of E one may define a Dirichlet series with coefficients in σE , convergent for $\text{Re}(s)$ sufficiently large:

$$L(\sigma, M, s) = \prod_p L_p(\sigma, M, s), \text{ where}$$

$$(2.2.1) \quad L_p(\sigma, M, s) = \sigma Z_p(M, p^{-s}), \text{ with } Z_p \in E(t) \subset E_\lambda(t), \text{ given by } Z_p(M, t) = \det(1 - F_p t, H_\lambda(M)^{I_p})^{-1} \text{ for } \lambda \nmid p.$$

As σ varies, these Dirichlet series are obtained from one another by conjugating their coefficients. We regard the system $L(\sigma, M, s)$ as a function $L^*(M, s)$ with values in the $\mathbb{C}$ -algebra $E \otimes \mathbb{C}$, identified with $\mathbb{C}^{\text{Hom}(E, \mathbb{C})}$ by

$$(2.2.2) \quad E \otimes \mathbb{C} \xrightarrow{\sim} \mathbb{C}^{\text{Hom}(E, \mathbb{C})}: e \otimes z \mapsto (z, \sigma(e))_\sigma.$$

This function can also be defined directly by an Euler product. As in 1.1, it is expected to admit analytic continuation in s .

The Euler product $L(\sigma, M, s)$ must be completed by an infinite factor $L_\infty(\sigma, M, s)$ depending on the Hodge realization of M . Setting $\Lambda(\sigma, M, s) = L_\infty(\sigma, M, s)L(\sigma, M, s)$, the conjectural functional equation is

$$(2.2.3) \quad \Lambda(\sigma, M, s) = \varepsilon(\sigma, M, s)\Lambda(\sigma, \check{M}, 1-s),$$

where $\varepsilon(\sigma, M, s)$, as a function of s , is the product of a constant and an exponential. The definitions of L_∞ and ε are recalled in 5.2. As above, the systems Λ and ε , for varying σ , are regarded as functions Λ^* and ε^* with values in $E \otimes \mathbb{C}$.

It follows from 2.5 below that $L_\infty(\sigma, M, s)$ is independent of σ , and that the infinite factor for the motive $R_{E/\mathbb{Q}}M$ obtained by restriction of the coefficient field is the $[E:\mathbb{Q}]$ -th power of $L_\infty(\sigma, M, s)$. This justifies the following.

PROPOSITION-DEFINITION 2.3. Let M be a motive over $\mathbb{Q}$ with coefficients in E . An integer n is critical for M if the following equivalent conditions hold: (i) n is critical for $R_{E/\mathbb{Q}}M$; (ii) neither $L_\infty(\sigma, M, s)$ nor $L_\infty(\sigma, \check{M}(1), s)$ has a pole at $s = n$. M is called critical if 0 is critical for M .

Our aim is to conjecture the value $L^*(M) =_{\text{def}} L^*(M, 0)$, for critical M , up to multiplication by an element of E . Equivalently, the values $L(\sigma, M) =_{\text{def}} L(\sigma, M, 0)$ are to be conjectured simultaneously, up to multiplication by a system of numbers $\sigma(e)$, with $e \in E$.

2.4. The Betti realization $H_B(M)$ is an E -vector space. Its dimension is the rank of M over E . The involution F_∞ is E -linear, so its $+$ and $-$ parts are E -vector subspaces. Their dimensions are denoted $d^+(M)$ and $d^-(M)$.

The complexification of $H_B(M)$ is a free $E \otimes \mathbb{C}$ -module. Identifying $E \otimes \mathbb{C}$ with $\mathbb{C}^{Hom(E, \mathbb{C})}$ by (2.2.2), one obtains a decomposition

$$H_B(M) \otimes \mathbb{C} = \bigoplus_{\sigma} H_B(\sigma, M),$$

where

$$H_B(\sigma, M) = (H_B(M) \otimes \mathbb{C}) \otimes_{E \otimes \mathbb{C}, \sigma} \mathbb{C},$$

or equivalently

$$H_B(\sigma, M) = H_B(M) \otimes_{E, \sigma} \mathbb{C}.$$

Since the Hodge summands $H_B^{pq}(M)$ are stable under E , each $H_B(\sigma, M)$ inherits a Hodge decomposition $H_B(\sigma, M) = \bigoplus H_B^{pq}(\sigma, M)$. The involution F_{∞} interchanges H^{pq} and H^{qp} ; in particular it stabilizes H_B^{pp} , which it splits into $+$ and $-$ parts. Let $h^{pq}(\sigma, M)$ be the dimension of $H_B^{pq}(\sigma, M)$, and $h^{pp\pm}(\sigma, M)$ that of $H_B^{pp\pm}(\sigma, M)$. The following proposition allows σ to be omitted from the notation.

PROPOSITION 2.5. The numbers $h^{pq}(\sigma, M)$ and $h^{pp\pm}(\sigma, M)$ are independent of σ .

We may and do assume that M is homogeneous. Then $H^{pq}(M)$ is the complexification of the E -vector space $Gr_F^p(H_{DR}(M))$; it is a free $E \otimes \mathbb{C}$ -module, proving the first assertion. For the second, $h^{pp\pm}(\sigma, M)$ is the excess of $d^{\pm}(M)$ over $\sum_{p>q} h^{pq}(\sigma, M)$.

2.6. Let M be a motive over $\mathbb{Q}$ with coefficients in E . For the remainder of this section, unless explicitly stated otherwise, assume that $R_{E/\mathbb{Q}}M$ satisfies the hypotheses of 1.7. The spaces $F^{\pm}$ and $H_{\overline{DR}}^{\pm}$ are now E -vector spaces. The isomorphisms (0.4.1) and (1.7.1),

$$\begin{aligned} (2.6.1) \quad & I: H_B(M) \otimes \mathbb{C} \rightarrow H_{DR}(M) \otimes \mathbb{C}, \\ (2.6.2) \quad & I^{\pm}: H_B^{\pm}(M) \otimes \mathbb{C} \rightarrow H_{\overline{DR}}^{\pm}(M) \otimes \mathbb{C}, \end{aligned}$$

are isomorphisms of $E \otimes \mathbb{C}$ -modules between complexifications of E -vector spaces. Set

$$c^{\pm}(M) = \det(I^{\pm}) \in (E \otimes \mathbb{C})^*, \quad \delta(M) = \det(I) \in (E \otimes \mathbb{C})^*,$$

where the determinant is computed in E -rational bases of $H_B^{\pm}(M)$ and $H_{\overline{DR}}^{\pm}(M)$, respectively $H_B(M)$ and $H_{DR}(M)$. The definition of $\delta(M)$ does not require the hypotheses imposed on M . Up to multiplication by an element of E^* , these numbers depend only on M . Again by 1.6, $c^+(M)$, $i^{d^-(M)}c^-(M)$, and $i^{d^-(M)}\delta(M)$ lie in $(E \otimes \mathbb{R})^*$.

Conjecture 2.7. Set $\Re(M) = -w/2$. If M is critical, (i) $L(\sigma, M, s)$ never has a pole at $s = 0$ and can vanish there only when $\Re(M) = 1/2$; (ii) the order of vanishing of $L(\sigma, M, s)$ at $s = 0$ is independent of σ .

For (i), see the discussion at the end of 1.3. B. Gross suggested that (ii) should be reasonable.

Conjecture 2.8. If M is critical and $L(\sigma, M) \neq 0$, then $L^*(M)$ is the product of $c^+(M)$ and an element of E^* .

REMARK 2.9. A motive M over a number field k with coefficients in E also defines an L -function $L^*(M, s)$. These functions are covered by the conjecture because $L^*(M, s) = L^*(R_{k/Q}M, s)$, where $R_{k/Q}$ denotes restriction of scalars from k to Q ; apply (0.5.3).

The functions $L(\sigma, M, s)$ are Euler products indexed by the finite places of k . They must be completed by the infinite factors $L_v(\sigma, M, s)$, indexed by the infinite places and defined in 5.2. In general they depend on σ . Only $L_\infty(\sigma, R_{k/Q}M, s)$, the product over the infinite places v of the $L_v(\sigma, M, s)$, is independent of σ .

REMARK 2.10. Let F be an extension of E , let $\iota: E \rightarrow F$ be the structural morphism, and let $\iota_C: E \otimes C \hookrightarrow F \otimes C$ be its complexification. Then $L^*(M \otimes_E F, s) = \iota_C L^*(M, s)$ and $c^\pm(M \otimes_E F) = \iota_C c^\pm(M)$. Thus the conjecture is compatible with extension of the coefficient field. If F/E is Galois with group G , Hilbert's Theorem 90, $H^1(G, F^*) = 0$, gives $((F \otimes C)^*/F^*)^G = (E \otimes C)^*/E^*$, so the conjecture is invariant under extension of coefficients.

REMARK 2.11. If E is an extension of F , then $L^*(R_{E/F}M, s) = N_{E/F}L^*(M, s)$. Since the periods $c^\pm$ satisfy the same identity, the conjecture is compatible with restriction of coefficients.

REMARK 2.12. Let D be a division algebra of rank d^2 over E . A motive M over Q with a D -module structure and rank n over D defines a Dirichlet series with coefficients in E whose Euler factors almost all have degree nd . For a finite place λ of E , $H_\lambda(M)$ is a free module over $D_\lambda = D \otimes_E E_\lambda$, and definition (2.2.1) is used with

$$Z_p(M, t) = \det \text{red}(1 - F_p t, H_\lambda(M)^{I_p})^{-1}.$$

If D_λ is a matrix algebra over E_λ and e is an indecomposable idempotent, the reduced determinant of an endomorphism A of a D_λ -module H is the determinant, over E_λ , of the restriction of A to eH ; the general case is obtained by descent.

The freedom afforded by 2.10 to extend the coefficient field also brings these L -functions under Conjecture 2.8. Choose an extension $\iota: E \hookrightarrow F$ that splits D and an indecomposable idempotent e of $D \otimes_E F$. Then $\iota_C L^*(M, s) = L^*(e(M \otimes_E F), s)$.

One may also define $c^+(M)$ directly in this setting, as explained in 2.13 below.

The remainder of this section is not used later.

2.13. Let D be a simple algebra over a field E , or even an Azumaya algebra over a ring. For the tensor formalism it is convenient to regard D -modules as “fictitious E -vector spaces”:

(a) To give an E -vector space V is equivalent to giving, for every étale extension F/E , an F -vector space V_F and compatible isomorphisms $V_G = V_F \otimes_F G$ for extensions G/F . One takes $V_F = V \otimes_E F$. By descent it is enough to give V_F for sufficiently large F .

(b) Let W be a D -module. For every étale extension F/E and every F -isomorphism $D \otimes F \cong \text{End}_F(L)$, with L free, set $W_{F,L} = \text{Hom}_{D \otimes F}(L, W \otimes F)$, where all unlabelled tensor products are over E . There is an isomorphism of $D \otimes F$ -modules

$$W \otimes_E F = L \otimes_F W_{F,L}.$$

If F is large enough to split D , L is unique up to a nonunique isomorphism; the nonuniqueness is due to scalar homotheties, which act trivially on $\text{End}(L)$. This is why the data $W_{F,L}$ are not of type (a); they form a “fictitious vector space over E .”

Let (W^α) be a family of D -modules, and let T be a tensor operation. If the scalar homotheties of L act trivially on $T(W_{F,L}^\alpha)$, the F -vector space $T(W_{F,L}^\alpha)$ is independent of the choice of L , and one obtains a system of type (a), hence an E -vector space $T(W^\alpha)$.

EXAMPLE. If W' and W'' are two D -modules of rank $n \cdot [D:E]^{1/2}$ over E , one may take $T = \text{Hom}(\wedge^n W'_{F,L}, \wedge^n W''_{F,L})$; one obtains a one-dimensional E -vector space $\delta(W', W'')$, and every homomorphism $f: W' \rightarrow W''$ has a reduced determinant $\det \text{red}(f) \in \delta(W', W'')$.

To define $c^+(M)$, apply this construction to the D -modules $H_B^+(M)$ and $H_{\overline{DR}}^+(M)$. Set $\delta = \delta(H_B^+(M), H_{\overline{DR}}^+(M))$. The reduced determinant of the isomorphism of $D \otimes \mathbb{C}$ -modules $I^+: H_B^+(M)_{\mathbb{C}} \rightarrow H_{\overline{DR}}^+(M)_{\mathbb{C}}$ lies in the free rank-one $E \otimes \mathbb{C}$ -module $\delta \otimes \mathbb{C}$. Set $\det \text{red}(I^+) = c^+(M) \cdot e$ for a basis e of δ .

3. Example: the ζ -function. 3.1. To understand the various realizations of the Tate motive $\mathbb{Z}(1)$, the simplest course is to write $\mathbb{Z}(1) = H_1(G_m)$. Since the multiplicative group is not a projective variety, this does not fit Grothendieck’s framework, which instead requires $\mathbb{Z}(1)$ to be defined as the dual of the direct factor $H^2(\mathbb{P}^1)$ of $H^*(\mathbb{P}^1)$.

The realization of $\mathbb{Z}(1)$ in $\mathbb{Z}_l$ -cohomology is the Tate module $T_l(G_m)$ of G_m :

$$\mathbb{Z}_l(1) = \text{proj} \lim \mu_{l^n}.$$

In the Hodge realization, $H_B(\mathbb{Z}(1)) = H_1(\mathbb{C}^*)$ is isomorphic to $\mathbb{Z}$ —or rather to $\mathbb{Q}$ in rational homology. This group is purely of type $(-1, -1)$, and $F_\infty = -1$.

In the de Rham realization, $H_{DR}(\mathbb{Z}(1))$ is the dual of $H_{DR}^1(G_m)$, isomorphic to $\mathbb{Q}$ and generated by the class of dz/z . The unique period of $H^1(G_m)$ is

$$(3.1.1) \oint dz/z = 2\pi i.$$

On $\mathbb{Z}_l(1)$, arithmetic Frobenius φ_p ($p \neq l$) acts by multiplication by p . Geometric Frobenius therefore acts by multiplication by p^{-1} ; this justifies the identity cited in 1.3:

$$(3.1.2) L(M(n), s) = L(M, n + s).$$

Since F_∞ acts as $(-1)^n$ on $H_B(\mathbb{Z}(n))$, $H_B^\varepsilon(\mathbb{Z}(n))$ vanishes for $\varepsilon = -(-1)^n$, and $c^\varepsilon(\mathbb{Z}(n)) = 1$. For $\varepsilon = (-1)^n$, (3.1.1) gives $c^\varepsilon(\mathbb{Z}(n)) = \delta(\mathbb{Z}(n)) = (2\pi i)^n$:

$$(3.1.3) c^\varepsilon(\mathbb{Z}(n)) = (2\pi i)^n \text{ for } \varepsilon = (-1)^n, c^\varepsilon(\mathbb{Z}(n)) = 1, \delta(\mathbb{Z}(n)) = (2\pi i)^n \text{ for } \varepsilon = -(-1)^n.$$

3.2. The function $\zeta(s)$ is the L -function attached to the unit motive $\mathbb{Z}(0) = H^*(\text{Point})$. The critical integers for $\mathbb{Z}(0)$ are the even integers > 0 and the odd integers ≤ 0 . Because $\zeta(s)$ has a pole at $s = 1$, 0 is not a trivial zero, and it would be reasonable to define the critical integers so as to include 0. Equation (3.1.3) and the known values

of $\zeta(n) = L(\mathbb{Z}(n))$ for critical n verify 1.8: $\zeta(n)$ is rational for odd $n \leq 0$, and is a rational multiple of $(2\pi i)^n$ for even $n \geq 0$.

4. Compatibility with the Birch and Swinnerton-Dyer conjecture. 4.1. Let A be an abelian variety over $\mathbb{Q}$, of dimension d . The Birch and Swinnerton-Dyer conjecture [15] asserts in particular: (a) $L(H^1(A), 1)$ is nonzero if and only if $A(\mathbb{Q})$ is finite. (b) Let ω be a generator of $H^0(A, \Omega^d)$. Then $L(H^1(A), 1)$ is the product of $\int_{A(\mathbb{R})} |\omega|$ and a rational number.

The motive $H^1(A)(1)$ is isomorphic to the dual $H_1(A)$ of $H^1(A)$; this expresses the existence of a polarization, an autoduality of $H^1(A)$ with values in $\mathbb{Z}(-1)$. By 1.7, $c^+(H^1(A)(1))$ is therefore computed as follows: if $\omega_1, \dots, \omega_d$ is a basis of $H^0(A, \Omega^1) = F^+H_{\text{DR}}^1(A)$, and $c_1, \dots, c_d$ is a basis of $H_1(A(\mathbb{C}), \mathbb{Q})^+$, then

$$(4.1.1) \quad c^+(H^1(A)(1)) = \det\langle \omega_i, e_j \rangle.$$

Again denoting by e_i a representative cycle, one has $\langle \omega_i, e_j \rangle = \int_{e_j} \omega_i$. Take as (e_i) a $\mathbb{Z}$ -basis of $H_1(A(\mathbb{R})^\circ, \mathbb{Z}) \subset H_1(A(\mathbb{C}), \mathbb{Z})$. The Pontryagin product of the e_i is represented by the d -cycle $A(\mathbb{R})^\circ$ in $A(\mathbb{C})$, with a suitable orientation; if ω is the exterior product of the ω_i , the determinant (4.1.1) is $\int_{A(\mathbb{R})^\circ} \omega$. Thus

$$\int_{A(\mathbb{R})} |\omega| = [A(\mathbb{R}) : A(\mathbb{R})^\circ] \left| \int_{A(\mathbb{R})^\circ} \omega \right|,$$

and 1.8 for $H^1(A)(1)$ is therefore equivalent to 4.1(b) above.

4.2. The Birch and Swinnerton-Dyer conjecture gives the exact value of $L(H^1(A), 1)$; the description of the rational factor in 4.1(b) in terms of the motive $H^1(A)(1)$ has the following form: (a) The datum $M = H^1(A)(1)$ is equivalent to that of A up to isogeny. One must begin by choosing A . This amounts to choosing an integral lattice in $H_B(M)$ whose l -adic completions are stable under $\text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$. (b) One then chooses ω , for example as the exterior product of the elements of a basis of an integral lattice in $H_{\text{DR}}(M)$. This ω determines the period $c^+(M)$ and, for each prime p , a rational factor $c_p(M)$. Almost all $c_p(M)$ equal 1, and the product formula ensures that $c^+(M) \cdot \prod_p c_p(M)$ is independent of ω . (c) Another rational number $h(M)$ is defined in terms of cohomological invariants of A ; the desired rational number is $h(M) \cdot \prod_p c_p(M)^{-1}$.

The invariance of the conjecture under isogeny is, moreover, a nontrivial theorem.

4.3. To generalize 4.1(a) to an arbitrary motive M of weight -1 , one would need an analogue of $A(\mathbb{Q})$. The group $A(\mathbb{Q})$ may be interpreted as the group of extensions of $\mathbb{Z}(0)$ by $H_1(A)$ in the category of 1-motives [6, §10] over $\mathbb{Q}$. This suggests considering the group of extensions of $\mathbb{Z}(0)$ by M in a category of mixed motives parallel to mixed Hodge structures, but not even a conjectural definition of such a category is available.

I shall only observe that, in all the usual cohomology theories, a cycle Y of dimension d , homologous to zero, on a proper algebraic variety

and smooth X determines a torsor under $H^{2d-1}(X)(d)$: there is an exact cohomology sequence

$$0 \rightarrow H^{2d-1}(X)(d) \rightarrow H^{2d-1}(X - Y)(d) \xrightarrow{-\partial} H_Y^{2d}(X)(d) \rightarrow H^{2d}(X)(d).$$

The cycle Y defines a cohomology class $\text{cl}(Y) \in H_Y^{2d}(X)(d)$ whose image in $H^{2d}(X)(d)$ is zero, and one takes $\partial^{-1}\text{cl}(Y)$. This construction is the analogue of assigning a point of the Jacobian to a degree-zero divisor on a curve.

5. Compatibility with the functional equation. Let E be a number field. In $(E \otimes \mathbb{C})^*$, write $\sim$ for the equivalence relation defined by the subgroup E^* .

PROPOSITION 5.1. Let M be a motive over $\mathbb{Q}$ with coefficients in E . Assume the hypotheses of 1.7. Then

$$c^+(M) \sim (2\pi i)^{-d^-(M)} \cdot \delta(M) \cdot c^+(\check{M}(1)).$$

For a free module L of rank n over a commutative ring A , where A is E or $E \otimes \mathbb{C}$, set $\det L = \wedge^n L$. By localization this definition extends to L projective of finite type; that generalization is not needed for the proof of 5.1. There are canonical isomorphisms

$$(5.1.1) \det(L^\vee) = \det(L)^{-1},$$

and, for any direct factor P of L ,

$$(5.1.2) \det(L) = \det(P) \cdot \det(L/P).$$

Here $\cdot$ denotes a tensor product and $^{-1}$ denotes a dual. Sign issues arise; they are best resolved by regarding $\det(L)$ as an invertible graded module placed in degree $\text{rank } L$. Since the final results are taken modulo $\sim$, we shall not be concerned with them.

For X and Y of the same rank, set $\delta(X, Y) = \text{Hom}(\det X, \det Y) = \det(X)^{-1} \cdot \det(Y)$. The determinant of $f: X \rightarrow Y$ lies in $\delta(X, Y)$. Equations (5.1.1) and (5.1.2) give isomorphisms

$$(5.1.3) \delta(X, Y) = \delta(Y^\vee, X^\vee),$$

and, for $F \subset X$ and $G \subset Y$,

$$(5.1.4) \delta(X, Y) = \delta(F, Y/G) \cdot \delta(G, X/F)^{-1}.$$

LEMMA 5.1.5. Under the isomorphism (5.1.3), $\det(u) = \det({}^t u)$.

LEMMA 5.1.6. Let F and G be direct factors of X and Y . Suppose that the isomorphism $f: X \rightarrow Y$ induces isomorphisms $f_F: F \rightarrow Y/G$ and $f_G^{-1}: G \rightarrow X/F$. Under the isomorphism (5.1.4), $\det(f) = \det(f_F) \det(f_G^{-1})^{-1}$.

The verification of these lemmas is left to the reader. For 5.1.6, note only that, with $G^1 = f^{-1}(G)$ and $F^1 = f(F)$, one has $X = F \oplus G^1$ and $Y = G \oplus F^1$, and f exchanges F with F^1 and G with G^1 .

Apply Lemma 5.1.6 to the complexifications of $H_B^+(M) \subset H_B(M)$ and $F^- \subset H_{\text{DR}}(M)$. Under the isomorphism (5.1.4), the determinant of $I: H_B(M) \otimes \mathbb{C} \rightarrow H_{\text{DR}}(M) \otimes \mathbb{C}$ is the product of the determinant of $I^+: H_B^+(M) \otimes \mathbb{C} \rightarrow$

$(H_{\text{DR}}(M)/F^-) \otimes \mathbb{C}$ and the inverse of the determinant of the morphism induced by Γ^{-1} :

$$J^- : F^- \otimes \mathbb{C} \rightarrow (H_B(M)/H_B^+(M)) \otimes \mathbb{C}.$$

The morphism J^- is the transpose of the morphism Γ^- for the dual motive $\check{M}$. Applying 5.1.5 and choosing E-bases of the spaces $\delta(X, Y)$ involved, one finally obtains

$$(5.1.7) \quad \delta(M) \sim c^+(M) \cdot c^-(\check{M})^{-1}.$$

Proposition 5.1 follows by applying to $\check{M}$ the following consequence of 3.1:

$$(5.1.8) \quad c^\pm(M) = (2\pi i)^{-d^\pm(M) \cdot n} c^{\pm(-1)^n}(M(n)).$$

For later use, also note the analogous formula

$$(5.1.9) \quad \delta(M) = (2\pi i)^{-d(M) \cdot n} \delta(M(n)).$$

5.2. We recall the exact form of the conjectural functional equation for motivic L-functions ([12], [4]). We work in the general case of a motive over a number field k with coefficients in a field E equipped with a complex embedding σ (cf. 2.9). First, the general form: (a) For each place v of k , define a local factor $L_v(\sigma, M, s)$. For finite v , the definition $L_v(\sigma, M, s) = \sigma Z_v(M, Nv^{-s})$, with $Z_v(M, t) \in E(t)$, depends on a compatibility hypothesis analogous to (1.2.1). One has $Z_v(M, t) = \det(1 - F_{v,t}, H_\lambda(M)^{I_v})^{-1}$. For an infinite place v induced by a complex embedding τ , $L_v(\sigma, M, s)$ is obtained by decomposing $H_\tau(M) \otimes_{E, \sigma} \mathbb{C}$ as a direct sum of minimal subspaces stable under the projectors defining the Hodge decomposition and, for real v , under F_∞ ; one associates to each subspace the “T-factor” in Table (5.3) and takes their product. For complex v , the minimal subspaces are one-dimensional, of type (p, q) . For real v , they are either two-dimensional of type $\{(p, q), (q, p)\}$, $p \neq q$, or one-dimensional of type (p, p) , with $F_\infty = \pm 1$. Write $\Lambda(\sigma, M, s)$ for the product of the $L_v(\sigma, M, s)$. (b) Let Ψ be a nontrivial character of the idele-class group $A \otimes k/k$, with components Ψ_v . For every place v of k , also choose a Haar measure dx_v on k_v . Assume that for almost every v , dx_v gives mass 1 to the integers of k_v , and that $\otimes dx_v$ is the Tamagawa measure, assigning mass 1 to the idele-class group.

Define local constants $\varepsilon_v(\sigma, M, s, \Psi_v, dx_v)$, almost all equal to 1 and each, as a function of s , a constant times an exponential. Let $\varepsilon(\sigma, M, s)$ be their product; it is independent of Ψ and the dx_v . (c) The conjectural functional equation is

$$\Lambda(\sigma, M, s) = \varepsilon(\sigma, M, s) \Lambda(\sigma, \check{M}, 1 - s).$$

Defining the ε_v requires an additional compatibility hypothesis among the $H_l(M)$. It assigns to v , σ , and M an isomorphism class of complex representations of the Weil group $W(\bar{k}_v/k_v)$ for infinite v , and of the enlarged Weil group $W'(\bar{k}_v/k_v)$ for finite v ; one takes the ε -factor of [4, 8.12.4], with $t = p^{-s}$.

For infinite v , this amounts to decomposing $H_\tau(M) \otimes_{E, \sigma} \mathbb{C}$ as in (a), and associating

a factor ε'_v to each subspace of the decomposition, then taking their product. The table of ε'_v for a particular choice of Ψ_v and dx_v is given in 5.3.

For finite v , first restrict the representations $H_l(M)$ to a decomposition group $\text{Gal}(\bar{k}_v/k_v) \subset \text{Gal}(\bar{k}/k)$, and then to the Weil group $W(\bar{k}_v/k_v)$. Applying [4, 8.3, 8.4] gives from $H_l(M)$ an isomorphism class ρ_l of representations of $W'(\bar{k}_v/k_v)$ over E_l . It is permissible, and useful, to replace it by its F-semisimplification [4, 8.6]. These representations, as l varies, are required to be compatible: after extending scalars from E_l to $\mathbb{C}$ by $\tilde{\sigma}: E_l \rightarrow \mathbb{C}$ extending σ , the isomorphism class obtained must be independent of l and σ . This is the required isomorphism class.

For the reader, we unpack the internal references to [4]. A representation of W' consists of a representation ρ of the Weil group in $\text{GL}(V)$, together with a nilpotent endomorphism N of V . In terms of the local constant of ρ in [4, 4.1], that of (ρ, N) is

$$\varepsilon((\rho, N), s, \Psi, dx) = \varepsilon(\rho \otimes \omega_s, \Psi dx) \cdot \det(-F N v^{-s}, V^{\rho(I)}/\text{Ker}(N)^{\rho(I)}).$$

REMARK 5.2.1. The function of s

$$\varepsilon((\rho, N), s, \Psi, dx) L((\rho, N)^\vee, 1-s) L((\rho, N), s)^{-1}$$

is the same for (ρ, N) and $(\rho, 0)$. Thus the conjectural functional equation for L-functions can be stated assuming compatibility only among the semisimplifications of the restrictions of $H_l(M)$ to a decomposition group.

5.3. The following table gives the local factors and constants attached to the various types of minimal subspaces in the Hodge realization. For the constants, Ψ_v and dx_v are chosen as follows: real v : $\Psi_v(x) = \exp(2\pi i x)$, measure dx ; complex v : $\Psi_v(z) = \exp(2\pi i \text{Tr}_{\mathbb{C}/\mathbb{R}}(z))$, measure $|dz \wedge d\bar{z}|$, so that for $z = x + iy$ these are $\exp(4\pi i x)$ and $2dx dy$.

Use the notation $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2} \Gamma(s/2)$, $\Gamma_{\mathbb{C}}(s) = 2 \cdot (2\pi)^{-s} \Gamma(s)$.

place type Γ -factor constant (for Ψ, dx above) complex (p, q) or (q, p) , $p \leq q$ $\Gamma_{\mathbb{C}}(s-p) i^{q-p}$ real $\{(p, q), (q, p)\}$, $p < q$ $\Gamma_{\mathbb{C}}(s-p) i^{q-p+1}$ (p, p) , $F_\infty = (-1)^{p+\varepsilon}$, $\varepsilon = 0$ or 1 $\Gamma_{\mathbb{R}}(s+\varepsilon-p) i^\varepsilon$

The case of interest here is $k = \mathbb{Q}$. One may then take $\Psi_\infty(x) = \exp(2\pi i x)$, $\Psi_p(x) = \exp(-2\pi i x)$, using the identification of $\mathbb{Q}_p/\mathbb{Z}_p$ with the p -primary part of $\mathbb{Q}/\mathbb{Z}$; dx_∞ is Lebesgue measure dx , and dx_p is Haar measure on $\mathbb{Q}_p$ giving $\mathbb{Z}_p$ mass 1.

PROPOSITION 5.4. If M is critical of weight w , then, modulo a rational number independent of σ ,

$$L_\infty(\sigma, \check{M}(1)) L_\infty(\sigma, M)^{-1} \sim (2\pi)^{-d^-(M)} \cdot (2\pi)^{-wd(M)/2}.$$

By 2.5, the $L_\infty(\sigma, \cdot)$ are independent of σ . It is therefore enough to verify 5.4 for a fixed σ . The formula is compatible with the substitution $M \mapsto \check{M}(1)$: $\check{M}(1)$ has

weight $-2 - w$, its d^- is $d^+(M)$, and abbreviating $d(M)$ and $d^\pm(M)$ as d and $d^\pm$, one has $d = d^+ + d^-$ and $(-d^- - wd/2) + (-d^+ - ((-2 - w)d)/2) = 0$.

It is therefore enough to verify 5.4 for $w \geq -1$. For an integer s , modulo $\mathbb{Q}^*$,

$$(5.4.1) \quad \Gamma_{\mathbb{R}}(s) \sim (2\pi)^{-s/2} \text{ for even } s > 0, \Gamma_{\mathbb{R}}(s) \sim (2\pi)^{(1-s)/2} \text{ for odd } s, \Gamma_{\mathbb{C}}(s) \sim (2\pi)^{-s} \text{ for } s > 0.$$

For $w \geq -1$, the power of 2π contributed by each subspace of $H_B(M) \otimes \mathbb{C}$, as in 5.2(a), is therefore

$$L_\infty(M) L_\infty(\check{M}(1)) L_\infty(\check{M}(1)) L_\infty(M)^{-1} \{(pq), (qp)\}, p \leq q \quad p-1-p-1-w \quad (pp), p \text{ even} \geq 0, F_\infty = -1 \quad p/2 -1-p/2 -1-w/2 \\ (pp), p \text{ odd} \geq 0, F_\infty = -1 \quad (1+p)/2 \quad (-1-p)/2 -1-w/2$$

The proposition follows immediately. Set $\det M = \wedge^{d(M)} M$, the exterior power over E .

PROPOSITION 5.5. $\varepsilon^*(M) \sim \varepsilon^*(\det M)$.

For the dx_v chosen as suggested in 5.4, we prove more precisely the equivalences

$$(5.5.1) \quad \varepsilon_v^*(M, \Psi_v, dx_v) \sim \varepsilon_v^*(\det M, \Psi_v, dx_v).$$

Set $\eta_v(\sigma, M, \Psi_v) = \varepsilon_v(\sigma, M, \Psi_v, dx_v) \varepsilon_v^{-1}(\sigma, \det M, \Psi_v, dx_v)$. This is equivalent to

$$(5.5.2) \quad \tau \eta_v(\sigma, M, \Psi_v) = \eta_v(\tau \sigma, M, \tau \Psi_v)$$

for every automorphism τ of $\mathbb{C}$.

It follows from [4, 5.4] that, if $a \in \mathbb{Q}_v^*$ has absolute value 1 and $(\Psi_v \cdot a)(x) = \Psi_v(ax)$, then

$$(5.5.3) \quad \eta_v(\sigma, M, \Psi_v) = \eta_v(\sigma, M, \Psi_v \cdot a) \text{ (for } \|a\|_v = 1).$$

For an automorphism τ of $\mathbb{C}$ and finite v , $\tau \Psi_v$ is of the form $\Psi_v \cdot a$ with $\|a\|_v = 1$; likewise, $\bar{\Psi}_\infty = \Psi_\infty \cdot (-1)$.

For finite v , the definition of ε_v is purely algebraic, so $\tau \eta_v(\sigma, M, \Psi_v) = \eta_v(\tau \sigma, M, \tau \Psi_v)$, and (5.5.2) follows from (5.5.3).

For $v = \infty$, one also has $\bar{\eta}_\infty(\sigma, M, \bar{\Psi}_\infty) = \eta_\infty(\bar{\sigma}, M, \bar{\Psi}_\infty) = \eta_\infty(\bar{\sigma}, M, \Psi_\infty)$. If Ψ_∞ is chosen as in 5.3, η_∞ is a power of i independent of σ ; hence $\eta_\infty(\sigma, M, \Psi_\infty) = \pm 1$, independently of σ , which proves (5.5.2).

THEOREM 5.6. Subject to Conjecture 6.6 on the nature of rank-one motives, Conjecture 2.8 is compatible with the conjectural functional equation of L-functions: one has

$$L_{\infty}^*(M)c^+(M) \sim \varepsilon^*(M)L_{\infty}^*(\check{M}(1))c^+(\check{M}(1)).$$

By 5.1, 5.3, and 5.5, this formula is equivalent to

$$(2\pi i)^{-d^-(M)} \cdot \delta(M) \sim (2\pi)^{-d^-(M)} \cdot (2\pi)^{-\text{wd}(M)/2} \cdot \varepsilon^*(\det M).$$

Set $D = \det M$ and $\varepsilon = d^-(D)$. Then $\delta(M) = \delta(D)$, $d^-(M) \equiv \varepsilon \pmod{2}$, and $\text{wd}(M)$ is the weight $w(D)$ of D , so that the formula is again equivalent to

$$(5.6.1) \quad \varepsilon^*(D) \sim (2\pi)^{w(D)/2} \cdot i^{\varepsilon} \cdot \delta(D).$$

In 6.5 we shall prove that (5.6.1) holds for a class of rank-one motives which conjecturally (6.6) includes them all.

6. Example: Artin L-functions. DEFINITION 6.1. The category of Artin motives is the Karoubian envelope of the dual of the category whose objects are zero-dimensional varieties over $\mathbb{Q}$ and whose morphisms are correspondences defined over $\mathbb{Q}$.

By definition, every zero-dimensional variety X defines an Artin motive $H(X)$; H is a fully faithful contravariant functor,

$$H: (\text{zero-dimensional varieties, correspondences}) \longrightarrow (\text{Artin motives}),$$

and every Artin motive is a direct factor of some $H(X)$.

6.2. Let us make this definition explicit. Let $\bar{\mathbb{Q}}$ be an algebraic closure of $\mathbb{Q}$, and let G be the Galois group $\text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$. A zero-dimensional variety is the spectrum of a finite product A of number fields, and Galois theory, in the form given to it by Grothendieck, says that the functor

$$X = \text{spec}(A) \mapsto X(\bar{\mathbb{Q}}) = \text{Hom}(A, \bar{\mathbb{Q}}):$$

(category of zero-dimensional varieties over $\mathbb{Q}$ and morphisms of schemes) $\longrightarrow$ (category of finite sets with a continuous G -action)

is an equivalence of categories. The inverse functor sends I to the spectrum of the ring of G -invariant functions from I to $\bar{\mathbb{Q}}$.

A correspondence from a zero-dimensional variety X to another Y is a formal rational linear combination of connected components of $X \times Y$. The map

$$\sum a_i Z_i \mapsto \sum a_i (\text{characteristic function of } Z_i(\bar{\mathbb{Q}}) \subset (X \times Y)(\bar{\mathbb{Q}}))$$

identifies correspondences with rational-valued G -invariant functions on $(X \times Y)(\bar{\mathbb{Q}}) = X(\bar{\mathbb{Q}}) \times Y(\bar{\mathbb{Q}})$, and composition of correspondences with matrix multiplication.

Let H be the contravariant functor sending X to the vector space $\mathbb{Q}^{X(\bar{\mathbb{Q}})}$ with its natural G -action; a correspondence $F: X \rightarrow Y$ is sent to the morphism $F^*: \mathbb{Q}^{Y(\bar{\mathbb{Q}})} \rightarrow \mathbb{Q}^{X(\bar{\mathbb{Q}})}$ whose matrix is ${}^t F$. This functor is fully faithful and identifies the category of Artin motives with that of rational representations of G .

6.3. In this model, if $\bar{\mathbb{Q}}$ is the algebraic closure of $\mathbb{Q}$ inside $\mathbb{C}$, the “Betti realization” functor H_B is the “underlying vector space” functor. The Hodge structure is purely of type $(0, 0)$, and the involution F_{∞} is the action of complex conjugation $F_{\infty} \in G$. Indeed, there is an isomorphism, functorial for correspondences,

$$H^*(X(\mathbb{C}), \mathbb{Q}) = \mathbb{Q}^{X(\mathbb{C})} = \mathbb{Q}^{X(\bar{\mathbb{Q}})} = H(X).$$

The “1-adic realization” functor H_l is $H_l(V) = V \otimes \mathbb{Q}_l$. Indeed, there is an isomorphism, functorial for correspondences,

$$H^*(X(\bar{\mathbb{Q}}), \mathbb{Q}_l) = \mathbb{Q}_l^{X(\bar{\mathbb{Q}})} = \mathbb{Q}^{X(\bar{\mathbb{Q}})} \otimes \mathbb{Q}_l.$$

Let us similarly compute the de Rham realization. For $X = \text{Spec}(A)$, $H_{\text{DR}}^*(X) = A$. Since $A = \text{Hom}_G(X(\bar{\mathbb{Q}}), \bar{\mathbb{Q}}) = (\mathbb{Q}^{X(\bar{\mathbb{Q}})} \otimes \bar{\mathbb{Q}})^G$, we obtain

$$(6.3.1) \quad H_{\text{DR}}(V) = (V \otimes \bar{\mathbb{Q}})^G.$$

Formula (6.3.1) realizes $H_{\text{DR}}(V)$ as a subspace of $V \otimes \bar{\mathbb{Q}}$. This subspace is a $\mathbb{Q}$ -structure: $(V \otimes \bar{\mathbb{Q}})^G \otimes \bar{\mathbb{Q}} \rightarrow \simeq V \otimes \bar{\mathbb{Q}}$. After extending scalars to $\bar{\mathbb{Q}}$, $H_B(V)$ and $H_{\text{DR}}(V)$ are therefore canonically isomorphic. Extending scalars further to $\mathbb{C}$ gives the isomorphism (0.4.1); this may be checked for $V = H(X)$.

6.4. Let E be a finite extension of $\mathbb{Q}$. The category of Artin motives with coefficients in E is obtained from the category of Artin motives as in 2.1. We identify it with the category of finite-dimensional E -vector spaces endowed with a G -action.

Rank-one Artin motives with coefficients in E correspond to characters $\varepsilon: G \rightarrow E^*$. We now compute their periods. Let $\varepsilon: G \rightarrow E^*$ and let f be its conductor; ε factors through a character, again denoted ε , of the quotient $(\mathbb{Z}/f\mathbb{Z})^* = \text{Gal}(\mathbb{Q}(\exp(2\pi i/f))/\mathbb{Q})$ of G . Let $[\varepsilon]$ denote the one-dimensional E -vector space E_ε on which G acts through ε . The Gauss sum

$$g = \sum \varepsilon(u) \otimes \exp(2\pi i u/f) \in [\varepsilon] \otimes \bar{\mathbb{Q}}$$

is nonzero and G -invariant; it is an E -basis of $H_{\text{DR}}([\varepsilon])$. The determinant of $I: H_B([\varepsilon]) \otimes \mathbb{C} \rightarrow \simeq H_{\text{DR}}([\varepsilon]) \otimes \mathbb{C}$, computed in the bases 1 and g , is g^{-1} . For every complex embedding σ of E one knows that $\sigma g \cdot \bar{\sigma} g = f$, a rational number independent of σ . Hence

$$(6.4.1) \quad \delta([\varepsilon]) \sim \sum \varepsilon^{-1}(u) \otimes \exp(-2\pi i u/f) \in (E \otimes \mathbb{C})^*.$$

PROPOSITION 6.5. Let D be the motive $[\varepsilon](n)$. It is a rank-one motive over $\mathbb{Q}$ with coefficients in E and weight $-2n$. Write $\varepsilon(-1) = (-1)^\eta$, with $\eta = 0$ or 1 . Then

$$\varepsilon^*(D) \sim (2\pi)^{-n} i^{\eta-n} \delta(D).$$

The constant in the functional equation of the Dirichlet L -function

$$L(\sigma, [\varepsilon], s) = \sum \sigma \varepsilon^{-1}(n) \cdot n^{-s},$$

where σ is a complex embedding of E , is known to be

$$\varepsilon(\sigma, [\varepsilon], s) = i^\eta \cdot f^s \cdot \sum \sigma \varepsilon(u)^{-1} \exp(-2\pi i u/f).$$

On the other hand, (5.1.9) gives $\delta(D) = (2\pi i)^n \delta([\varepsilon])$. Applying (6.4.1), and observing that for integral $s = n$ the number f^n is rational and independent of σ , proves the proposition.

This proposition verifies (5.6.1) for the motives $[\varepsilon](n)$. To complete the proof of 5.6, it remains only to state:

Conjecture 6.6. Every rank-one motive over $\mathbb{Q}$ with coefficients in E is of the form $[\varepsilon](n)$, where ε is a character of $G = \text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ with values among the roots of unity in E , and n is an integer.

PROPOSITION 6.7. Conjecture 2.8 is true for Artin L-functions.

For Artin motives, the functional equation of the L-functions is available. Moreover, the determinant of an Artin motive, after a Tate twist, is of the type predicted in 6.6. The arguments of Sections 5 and 6 therefore show the compatibility of 2.8 with the functional equation, and it is enough to prove 2.8 for motives $V(n)$, where V is an Artin motive and $n \leq 0$ is an integer. If $V(n)$ is critical, F_∞ acts on V by multiplication by $-(-1)^n$ and $c^+ = 1$ (cf. 3.1); one must prove that for every embedding $\sigma: E \rightarrow \mathbb{C}$ and every automorphism τ of $\mathbb{C}$,

$$\tau L(\sigma, V(n)) = L(\tau\sigma, V(n)).$$

This follows from Siegel's results [14]. See [2, 1.2].

7. L-functions attached to modular forms. 7.1. Put $q = e^{2\pi iz}$, and let $f = \sum a_{nq}^n$ be a primitive holomorphic cuspidal modular form (new form) of weight $k \geq 2$, conductor N , and character ε . The Dirichlet series $\sum a_{nn}^{-s}$ has an Euler-product expansion whose local factor at $p \nmid N$ is

$$(1 - a_{pp}^{-s} + \varepsilon(p)p^{k-1}p^{-2s})^{-1}.$$

Let E be the subfield of $\mathbb{C}$ generated by the a_n . The form f should give rise to a rank-two motive $M(f)$ with coefficients in E , of Hodge type $\{(k-1, 0), (0, k-1)\}$, determinant $[\varepsilon^{-1}](1-k)$ in the notation of 6.4, and L-function given by the Dirichlet series $\sum a_{nn}^{-s}$.

I have not tried to define $M(f)$ as a motive in Grothendieck's sense. One difficulty is that $M(f)$ appears naturally as a direct factor in the cohomology of a noncompact variety, or as a direct factor in the cohomology of a complete modular curve with coefficients in the direct image of a locally constant sheaf—more properly, a local system of motives—on the finite part of that modular curve. This lies outside Grothendieck's formalism, but it permits the realizations of $M(f)$ to be defined.

7.2. First suppose that $k = 2$ and ε is trivial. Let X be the Poincaré upper half-plane, let N be the conductor of f , and let $\Gamma_0(N)$ be the subgroup of $SL(2, \mathbb{Z})$ consisting of matrices whose reduction modulo N has the form $\begin{pmatrix} * & * \\ 0 & * \end{pmatrix}$. Put

$$\omega_f = \sum a_{nq}^n \cdot dq/q = \sum a_{nq}^n \cdot 2\pi i \, dz.$$

The form ω_f is a holomorphic differential on the completed curve $\bar{M}(\Gamma_0(N))$ associated with $M(\Gamma_0(N)) = X/\Gamma_0(N)$. It is an eigenvector for the Hecke correspondences:

$$(7.2.1) \quad T_n^* \omega_f = a_n \omega_f \quad (\text{for } n \text{ prime to } N),$$

and is characterized up to a factor by (7.2.1). This follows from strong multiplicity one and the theory of primitive forms. The reader may instead supplement (7.2.1) by the analogous condition for n not prime to N , and do the same below; the assertion then becomes elementary, since the completed condition (7.2.1) determines the Taylor expansion of ω_f at the cusp $i\infty$ up to a factor.

Abbreviate $M(\Gamma_0(N))$ to M and $\bar{M}(\Gamma_0(N))$ to $\bar{M}$. These curves have a natural $\mathbb{Q}$ -structure for which the Hecke correspondences are defined over $\mathbb{Q}$. The form ω_f is defined over E , and its conjugate by an automorphism σ of $\mathbb{C}$ is $\omega_{\sigma f}$, obtained by applying σ to the coefficients.

In the language of 2.1(B), define $M(f)$ to be the submotive of $H^1(\bar{M})_E$ that is the kernel of the endomorphisms $T_n^* - a_n$ —if one wishes to use only

kernels of projectors, replace $T_n^* - a_n$ by $P(T_n - a_n \Delta)^*$ for a suitable polynomial P . The motive $M(f)$ has the properties stated in 7.1.

In each cohomology theory $\mathcal{H}$ under consideration, the map

$$(7.2.2) \quad \mathcal{H}_c^1(M) \longrightarrow \mathcal{H}^1(\bar{M})$$

is surjective, and the systems of eigenvalues of the T_n^* on its kernel do not occur in $\mathcal{H}^1(\bar{M})$. The $\mathcal{H}$ -realization of $M(f)$ is therefore still the common kernel, inside $\mathcal{H}_c^1(M) \otimes E$, of the $T_n^* - a_n$.

The cohomology group $H_c^1(M, \mathbb{Q})$ carries a mixed Hodge structure of type $\{(0, 0), (0, 1), (1, 0)\}$, with $H^1(\bar{M}, \mathbb{Q})$ as its quotient of type $\{(0, 1), (1, 0)\}$. In particular, (7.2.2) induces an isomorphism on the F^1 -subspaces of the Hodge filtration; every holomorphic differential ω on $\bar{M}$ therefore defines a compactly supported cohomology class on M . This can be seen directly. If $\mathcal{J} \subset \mathcal{O}$ is the ideal of the cusps, de Rham cohomology $H_c^*(M)$ is the hypercohomology on $\bar{M}$ of the complex $\mathcal{J} \rightarrow \Omega^1$; analytically this complex resolves the constant sheaf $\mathbb{C}$ on M extended by zero to $\bar{M}$, and H^1 receives $H^0(\bar{M}, \Omega^1)$.

For simplicity suppose $E = \mathbb{Q}$, and compute $c^+(M(f)(1))$. The motive $M(f)(1)$ is dual to $M(f)$, so c^+ is a period of $M(f)$ by 1.7: one integrates ω_f against a rational homology class of $\bar{M}$ fixed by F_∞ . Lifting $M(f)$ to $\mathcal{H}_c^1(M)$ shows that one may instead integrate ω_f against a homology class without compact support on M fixed by F_∞ . All nonzero integrals of this type are commensurable.

To compute F_∞ , it is useful to write M as the quotient of $X^\pm = \mathbb{C} - \mathbb{R}$ by the subgroup of $GL(2, \mathbb{Z})$ whose reduction modulo N has the form $((*, *), (0, *))$. Complex conjugation is induced by $z \mapsto \bar{z}$, and the image of $i\mathbb{R}^+$ in $M(\mathbb{C})$ is a noncompact cycle fixed by F_∞ . The formula

$$L(M(f), 1) = -\int_0^{i\infty} \omega_f$$

therefore verifies 1.8 for $M(f)(1)$: the right-hand side is either zero or equivalent to $c^+(M(f)(1))$.

REMARK 7.3. One also has

$$c^\pm(M(f)(1)) \sim \int_{a/b}^{i\infty} \omega_f \pm \int_{-a/b}^{i\infty} \omega_f.$$

7.4. For arbitrary ε and E , $\Gamma_0(N)$ must be replaced by $\Gamma_1(N)$, the subgroup of $SL(2, \mathbb{Z})$ consisting of matrices whose reduction modulo N has the form $((1, *), (0, 1))$. As above, for computing F_∞ it is more convenient to work with $X^\pm$ and the subgroup of $GL(2, \mathbb{Z})$ whose reduction modulo N has the form $((1, *), (0, *))$. A dualization also appears, hidden in 7.2 by the symmetry of the correspondence T_n . For a suitable definition of T_n one has:

(a) $M(f)$ is the common kernel of the $T_n^* - a_n$ in $H^1(\bar{M})$, where $\bar{M} = X/\Gamma_1(N)$.

(b) One has ${}^t T_n^* \omega_f = a_n \omega_f$ and $T_n^* \omega_f = \bar{a}_n \omega_f$, where $\bar{a}_n = \varepsilon(n)^{-1} a_n$. Thus ω_f belongs to the de Rham realization of $M(f) = M(f)^\vee(-1)$.

It follows that $c^+(M(f)(1)) \in (E \otimes \mathbb{C})^*/E^* \cong \mathbb{C}^{Hom(E, \mathbb{C})}/E^*$ is given by the system of periods

$$c^+(M(f)(1)) \sim (\int_0^{i\infty} \omega_{\sigma f})_\sigma,$$

provided this system is nonzero. If one of these integrals vanishes, they all vanish. This happens exactly when, in the part of noncompact homology

spanned by the cycle $i\mathbb{R}^+$ and its transforms under the T_n , the system of eigenvalues a_n for the T_n does not occur. This proves 2.8, and partly 2.7, for $M(f)(1)$.

REMARK 7.5. The systems of T_n -eigenvalues in $\text{Ker}(\mathcal{H}_c^1(M) \rightarrow \mathcal{H}^1(\bar{M}))$ are associated with Eisenstein series, whereas those occurring in $\mathcal{H}^1(\bar{M})$ are associated with cusp forms. The two sets are therefore disjoint. It follows that the mixed Hodge structure on $H_c^1(M, \mathbb{Q})$ is a direct sum of Hodge structures: the extension of $H^1(\bar{M}, \mathbb{Q})$ by $\text{Ker}(H_c^1(M, \mathbb{Q}) \rightarrow H^1(\bar{M}, \mathbb{Q}))$ splits. Generalized to any congruence subgroup of $\text{SL}(2, \mathbb{Z})$, this is equivalent to Manin's theorem [9] that the difference of two cusps always has finite order in the Jacobian (cf. [6, 10.3.4, 10.3.8, and 10.1.3]). The proof just given does not differ in substance from Manin's.

7.6. For arbitrary weight k , the cohomology of $\bar{M}$ must be replaced by cohomology with coefficients in a suitable sheaf. To describe the construction, replace M and $\bar{M}$ by M_n and $\bar{M}_n$ for the congruence subgroup $\Gamma(n)$, where n is a multiple of N and $n \geq 3$. One then descends to M and $\bar{M}$ by taking invariants under a suitable finite group. Let $g: \mathcal{E} \rightarrow M_n$ be the universal elliptic curve and j the inclusion $M_n \rightarrow \bar{M}_n$. The relevant cohomology is

$$H^1(\bar{M}_n, j_* \text{Sym}^{k-2}(R^1 g_* \mathbb{Q})).$$

The realizations of $M(f)$ are direct factors of this group, computed in the corresponding cohomology theory. As before they may be lifted to

$$H_c^1(M_n, \text{Sym}^{k-2}(R^1 g_* \mathbb{Q})).$$

To compute $c^\pm$ of the dual of $M(f)$, one integrates the class in H_c^1 defined by f against noncompact homology classes on M_n with coefficients in the local system dual to $\text{Sym}^{k-2}(R^1 g_* \mathbb{Q})$. Taking the cycle that is the image of $i\mathbb{R}^+$, together with various sections of the dual local system forming a basis, gives the Eichler integrals

$$\int_0^{i\infty} f(q) \cdot dq/q \cdot (2\pi iz)^l \quad (0 \leq l \leq k-2),$$

even periods for even l and odd periods for odd l . Critical values $L(M(f), n)$ can be expressed in terms of these periods.

PROPOSITION 7.7. Let M be a rank-two motive with coefficients in E and Hodge type $\{(a, b), (b, a)\}$, where $a \neq b$. Then $d^\pm \text{Sym}^n(M)$ and $c^\pm \text{Sym}^{nM}$ are given by the following formulas:

(1) if $n = 2l + 1$, then $d^\pm = l + 1$ and

$$c^\pm \text{Sym}^{nM} = c^\pm(M)^{(l+1)(l+2)/2} c^\mp(M)^{l(l+1)/2} \delta(M)^{l(l+1)/2};$$

(2) if $n = 2l$, then $d^+ = l + 1$, $d^- = l$, and

$$c^+ \text{Sym}^{nM} = (c^+(M)c^-(M))^{l(l+1)/2} \delta(M)^{l(l+1)/2}, \quad c^- \text{Sym}^{nM} = (c^+(M)c^-(M))^{l(l+1)/2} \delta(M)^{l(l-1)/2}.$$

This is a question of elementary linear algebra, which we treat by "dimensional analysis."

(a) $H_B(M)$ is a rank-two E -vector space with an involution F_∞ having matrix $((0, 1), (1, 0))$ in a suitable basis e^+, e^- . The quantities $d^\pm \text{Sym}^{nM}$ are the dimensions of the plus and minus parts of the n th symmetric power, with respective bases

$$\{(e^+)^n, (e^+)^{n-2}(e^-)^2, \dots\} \text{ and } \{(e^+)^{n-1}e^-, (e^+)^{n-3}(e^-)^3, \dots\}.$$

The stated dimensions follow.

(b) Let ω, η be a basis of the dual of $H_{DR}(M)$, with ω annihilating $F^+H_{DR}(M)$. The subspace $F^\pm$ of $H_{DR}(\text{Sym}^{nM})^\vee \sim \text{Sym}^{nH}_{DR}(M)^\vee$ has basis

$$\omega^n, \omega^{n-1}\eta, \dots, \omega^{n-d^\pm+1}\eta^{d^\pm-1},$$

and

$$(7.7.1)^+ c^+ \text{Sym}^{nM} = \langle \omega^n \wedge (\omega^{n-1}\eta) \wedge \dots, (e^+)^n \wedge ((e^+)^{n-2}(e^-)^2) \wedge \dots \rangle,$$

$$(7.7.1)^- c^- \text{Sym}^{nM} = \langle \omega^n \wedge (\omega^{n-1}\eta) \wedge \dots, ((e^+)^{n-1}e^-) \wedge ((e^+)^{n-3}(e^-)^3) \wedge \dots \rangle.$$

(c) Put $V = H_B(M) \otimes \mathbb{C} \sim H_{DR}(M) \otimes \mathbb{C}$; this is an $E \otimes \mathbb{C}$ -module. The formulas above show that $c^\pm \text{Sym}^{nM}$ depends only on V as an $E \otimes \mathbb{C}$ -module, on its basis e^+, e^- , on $\omega \in V^*$, and on the image $\bar{\eta}$ of η in $V^*/\langle \omega \rangle$. The $d^\pm$ -vector on the left of the pairing in (7.7.1) does not change when η is replaced by $\eta + \lambda\omega$. Moreover, $\langle \omega, e^+ \rangle$ and $\langle \omega, e^- \rangle$ are invertible and $\bar{\eta}$ is a basis of $V^*/\langle \omega \rangle$. Up to isomorphism, the system $(V, e^+, e^-, \omega, \bar{\eta})$ is therefore described by

$$c^+ = \langle \omega, e^+ \rangle, c^- = \langle \omega, e^- \rangle, \delta = \langle \omega \wedge \eta, e^+ \wedge e^- \rangle$$

in $(E \otimes \mathbb{C})^*$. Replacing $e^+, e^-, \omega, \bar{\eta}$ by $\lambda e^+, \mu e^-, \alpha\omega, v\bar{\eta}/\lambda\mu$ replaces c^+, c^- , and δ by $\lambda\alpha c^+, \mu\alpha c^-$, and $v\delta$. When $c^+ = c^- = \delta = 1$, the right-hand side of (7.7.1) $^\pm$ lies in $\mathbb{Q}^*$. In general, $c^\pm \text{Sym}^{nM}$ is therefore a rational multiple of the product of $c^+(M)$, $c^-(M)$, and $\delta(M)/c^+(M)c^-(M)$, raised to the respective degrees with which e^+, e^- , and $\bar{\eta}$ occur in (7.7.1). Half the calculation may be avoided by observing that replacing F_∞ by $-F_\infty$ exchanges e^+ and e^- , hence c^+ and c^- , preserves δ , and exchanges (7.7.1) $^+$ and (7.7.1) $^-$ when n is odd while preserving them when n is even.

For $n = 2l + 1$, $d^+ = d^- = l + 1$:

$\deg \bar{\eta}$ in (7.7.1) $^\pm = 0 + 1 + \dots + l = l(l + 1)/2$, $\deg e^+$ in (7.7.1) $^+ = (2l + 1) + (2l - 1) + \dots + 1 = (l + 1)^2 = \deg e^-$ in (7.7.1) $^-$, $\deg e^+$ in (7.7.1) $^- = 2l + (2l - 2) + \dots + 0 = l(l + 1) = \deg e^-$ in (7.7.1) $^+$.

For $n = 2l$, $d^+ = l + 1$ and $d^- = l$:

$\deg \bar{\eta}$ in (7.7.1) $^+ = 0 + 1 + \dots + l = l(l + 1)/2$, $\deg \bar{\eta}$ in (7.7.1) $^- = 0 + 1 + \dots + (l - 1) = l(l - 1)/2$, $\deg e^\pm$ in (7.7.1) $^+ = 2l + (2l - 2) + \dots + 0 = l(l + 1)$, $\deg e^\pm$ in (7.7.1) $^- = (2l - 1) + (2l - 3) + \dots + 1 = l^2$.

7.8. This proposition gives a conjecture for the values at critical integers of $L(\text{Sym}^{nM}(f), s)$. The formulas are as follows:

$$(7.8.1) L(\sigma, M(f), s) = \sum \sigma a_n n^{-s} = \prod_p L_p(\sigma, M(f), s),$$

where for almost every p ,

$$L_p(M(f), s) = (1 - a_{pp}^{-s} + \varepsilon(p)p^{k-1-2s})^{-1} = ((1 - \alpha'_{pp}^{-s})(1 - \alpha''_{pp}^{-s}))^{-1}.$$

Here ε is a Dirichlet character. Since $\wedge^2 M(f) = [\varepsilon^{-1}](1 - k)$, one obtains

$$(7.8.2) \delta(M(f)) \sim (2\pi i)^{1-k} \cdot \sum_{u \bmod F} \varepsilon(u) \otimes \exp(-2\pi i u/F),$$

if ε has conductor F , and

$$(7.8.3) L^*(M(f), m) \sim (2\pi i)^m c^\pm(M(f)), \pm = (-1)^m, \text{ for } 1 \leq m \leq k - 1.$$

$$(7.8.4) \quad L(\sigma, \text{Sym}^{nM}(f), s) = \prod_p L_p(\sigma, \text{Sym}^{nM}(f), s),$$

where for almost every p ,

$$L_p(\text{Sym}^{nM}(f), s)^{-1} = \prod_{i=0}^n (1 - \alpha'_p \alpha''_p \beta_p^{n-i} p^{-s}).$$

Conjecturally, for critical m ,

$$L^*(\text{Sym}^{nM}(f), m) \sim (2\pi i)^{m d^{\pm} \text{Sym}^n M(f)} \cdot c^{\pm} \text{Sym}^{nM}(f), \pm = (-1)^m,$$

where $c^{\pm}$ and $d^{\pm}$ are given in terms of $c^{\pm}(M)$ and $\delta(M)$, characterized by (7.8.2) and (7.8.3), through the formulas of 7.7.

For numerical evidence supporting this conjecture, see [18].

8. Algebraic Hecke characters. In [3, paragraph 5], whose notation we shall use, the reader will find the essential definitions concerning algebraic Hecke characters (= Grössencharaktere of type A_0).

Conjecture 8.1. Let k and E be two finite extensions of $\mathbb{Q}$, and let χ be an algebraic Hecke character of k with values in E .

(i) There exists a rank-one motive $M(\chi)$ over k , with coefficients in E , such that, for every place λ of E , the λ -adic representation $H_{\lambda}M(\chi)$ is the one defined by χ : geometric Frobenius at a prime $\mathfrak{p}$ prime to the conductor of χ and to the residue characteristic l of λ acts by multiplication by $\chi(\mathfrak{p})$.

(ii) This motive is characterized up to isomorphism by this property.

(iii) Every rank-one motive is of the form $M(\chi)$.

(iv) Decompose $k \otimes E$ as a product of fields, $k \otimes E = \prod K_i$, and write the algebraic part $\chi_{\text{alg}}: k^* \rightarrow E^*$ of χ in the form

$$\chi_{\text{alg}}(x) = \prod N_{K_i/E}(x)^{n_i}.$$

The decomposition of $k \otimes E$ induces a decomposition of $H_{\text{DR}}(M(\chi))$ into

$$H_{\text{DR}}(M(\chi))_i := H_{\text{DR}}(M(\chi)) \otimes_{k \otimes E} K_i;$$

with this notation, the Hodge filtration is the filtration by

$$\oplus_{n_i \geq p} H_{\text{DR}}(M(\chi))_i.$$

The uniqueness in 8.1(ii) imposes the following formalism on the $M(\chi)$:

$$(8.1.1) \quad M(\chi' \chi'') \sim M(\chi') \otimes M(\chi'').$$

(8.1.2) If $\iota: E \rightarrow E'$ is a finite extension of E , then $M(\iota\chi)$ is obtained from $M(\chi)$ by extension of coefficients from E to E' .

(8.1.3) If k' is a finite extension of k , then $M(\chi \circ N_{k'/k})$ is obtained from $M(\chi)$ by extension of scalars from k to k' .

REMARK 8.2. Introduce the notation c = complex conjugation, $\bar{\mathbb{Q}}$ = the algebraic closure of $\mathbb{Q}$ in $\mathbb{C}$, $S = \text{Hom}(k, \bar{\mathbb{Q}}) = \text{Hom}(k, \mathbb{C})$, and $J = \text{Hom}(E, \bar{\mathbb{Q}}) = \text{Hom}(E, \mathbb{C})$. The decomposition of $k \otimes E$ into the fields K_i corresponds to the partition of $S \times J$ into $\text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ -orbits.

Every algebraic homomorphism $\eta: k^* \rightarrow E^*$ can be written in the form $\prod N_{K_i/E}(x)^{n_i}$. If the $\text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ -orbit of $(\sigma, \tau) \in S \times J$ corresponds to K_i , that is, if $\sigma \otimes \tau: k \otimes E \rightarrow \mathbb{C}$ factors through K_i , write $n(\eta; \sigma, \tau)$, or simply $n(\sigma, \tau)$, for the integer n_i . The function $n(\sigma, \tau)$ is constant on $\text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ -orbits. If η is the algebraic part of a Hecke character χ , then in addition

$$(8.2.1) \quad \text{The integer } w = n(\eta; \sigma, \tau) + n(\eta; c\sigma, \tau) \text{ is independent of } \sigma \text{ and } \tau.$$

This is the weight of χ , and of $M(\chi)$. We shall sometimes write $n(\chi; \sigma, \tau)$ for $n(\chi_{\text{alg}}; \sigma, \tau)$.

Conversely, a homomorphism η satisfying (8.2.1) is nearly of the form χ_{alg} for a suitable Hecke character χ : (a) some power of it is of this form; (b) there is a finite extension $\iota: E \rightarrow E'$ of E such that $\iota\eta$ is of this form; (c) there is a finite extension k' of k such that $\eta \circ N_{k'/k}$ is of this form.

Rule 8.1(iv) determines the Hodge bigrading of $H_\sigma M(\chi)$ from its E -module structure: the direct factor $H_\sigma M(\chi) \otimes_{E, \tau} \mathbb{C}$ of $H_\sigma M(\chi) \otimes \mathbb{C}$ has Hodge type (p, q) , where $p = n(\chi; \sigma, \tau)$ and $q = w - p = n(\chi; \sigma, c\tau)$.

EXAMPLE 8.3. Let A be an abelian variety over k with complex multiplication by E . Suppose $H_1(A)$ has rank one over E (CM type). It follows from the theory of Shimura and Taniyama that $H_1(A)$ satisfies condition 8.1(i) for an algebraic Hecke character χ of k with values in E , and that the algebraic part χ_{alg} of χ is read from the $k \otimes E$ -module $\text{Lie}(A)$:

$$\chi_{\text{alg}}(x) = \det_E(x \otimes 1, \text{Lie}(A))^{-1}.$$

EXAMPLE 8.4. Let k be the field of n -th roots of unity, let V be the Fermat hypersurface with projective equation $\sum_{i=0}^m X_i^n = 0$, and let M be the motive “primitive cohomology in the middle dimension of V^m .” Let G be the quotient of μ_n^{m+1} by its diagonal subgroup. This group acts on V by $(\alpha_i) \cdot (X_i) = (\alpha_i X_i)$, and on M by transport of structure. Decomposing M by means of the product decomposition of the group algebra $\mathbb{Q}[G]$, one obtains motives satisfying 8.1(i) for suitable algebraic Hecke characters, namely those introduced by Weil in his study of Jacobi sums.

EXAMPLE 8.5. The motive $\mathbb{Z}(-1)$ satisfies 8.1(i) for χ equal to the norm. For χ of finite order, a suitable Artin motive satisfies 8.1(i).

8.6. For every $\sigma \in S$, $H_\sigma(M(\chi))$ has rank one over E . Choose a basis e_σ of each $H_\sigma(M(\chi))$. Besides its E -module structure, the sum of the $H_\sigma(M(\chi)) \otimes \mathbb{C}$ has a natural $k \otimes \mathbb{C} = \mathbb{C}^S$ -module structure; altogether it is a free rank-one $k \otimes E \otimes \mathbb{C}$ -module, for which $e = \sum e_\sigma$ is a basis.

8.7. The de Rham realization $H_{\text{DR}}(M(\chi))$ is a free rank-one $k \otimes E$ -module. Choose a basis ω . The sum of the maps $I_\sigma: H_\sigma(M(\chi)) \otimes \mathbb{C} \xrightarrow{\sim} H_{\text{DR}}(M(\chi)) \otimes_{k, \sigma} \mathbb{C}$ is the isomorphism (0.4) of $k \otimes E \otimes \mathbb{C}$ -modules

$$I: \oplus_\sigma H_\sigma(M(\chi)) \otimes \mathbb{C} \xrightarrow{\sim} H_{\text{DR}}(M(\chi)) \otimes_{\mathbb{Q}} \mathbb{C}.$$

Put $p'(\chi) = \omega/I(e) \in (k \otimes E \otimes \mathbb{C})^*$. This period depends on χ , ω , and e . Modulo $(k \otimes E)^*$ and E^{*S} , it depends only on χ . Write $p'(\chi; \sigma, \tau)$, or simply $p'(\sigma, \tau)$, for the component indexed by (σ, τ) of its image under the isomorphism $k \otimes E \otimes \mathbb{C} \xrightarrow{\sim} \mathbb{C}^{S \times J}$.

8.8. Let A be as in 8.3 and calculate the periods $p'(\sigma, \tau)$, modulo $\bar{\mathbb{Q}}^*$, for the motive $H^1(A)$. First extend scalars from k to $\bar{\mathbb{Q}}$ using σ . If $n(\sigma, \tau) = 1$, there is a holomorphic 1-form ω defined over $\bar{\mathbb{Q}}$ such that $u^*\omega = \tau(u)\omega$ for $u \in E$, and, for $Z \in H_1(A(\mathbb{C}))$, one has $p'(\sigma, \tau) \sim \int_Z \omega$. The general case follows from

$$p'(\sigma, \tau) \cdot p'(\sigma, c\tau) \sim 2\pi i.$$

8.9. Conjecture 8.1(ii) asserts in particular that two motives satisfying 8.1(i) have the same period p . For the motives of 8.4, the periods are expressed

in terms of values of the Γ function and, when working modulo $\tilde{\mathbb{Q}}^*$, 8.1 suggests B. Gross's conjecture [7] relating certain periods to products of Γ -values.

Comparison of 8.4 and 8.5 leads to Conjectures 8.11 and 8.13 below. The result announced after 0.10 proves them.

8.10. Let N be an integer, $k = \mathbb{Q}(\exp(2\pi i/N))$, $\mathfrak{P}$ a prime ideal of k prime to N , $k_{\mathfrak{P}}$ its residue field, and $q = N\mathfrak{P} = |k_{\mathfrak{P}}|$. Let t denote the inverse of reduction modulo $\mathfrak{P}$, from the N -th roots of unity in $k_{\mathfrak{P}}$ to those in k .

For $a \in N^{-1}\mathbb{Z}/\mathbb{Z}$, $a \neq 0$, consider the Gauss sum

$$g(\mathfrak{P}, a, \Psi) = - \sum_{x \in k_{\mathfrak{P}}^*} t(x^{-a(q-1)}) \Psi(x).$$

The sum is over $k_{\mathfrak{P}}^*$, and $\Psi: k_{\mathfrak{P}} \rightarrow \mathbb{C}^*$ is a nontrivial additive character.

Let $\mathbf{a} = \sum n(a)\delta_a$ in the free abelian group with basis $N^{-1}\mathbb{Z}/\mathbb{Z} - \{0\}$. If $\sum n(a)a = 0$, the product of the $g(\mathfrak{P}, a, \Psi)^{n(a)}$ is independent of Ψ , and set

$$g(\mathfrak{P}, \mathbf{a}) = \prod_a g(\mathfrak{P}, a, \Psi)^{n(a)}.$$

Weil [16] proved that, as a function of $\mathfrak{P}$, $g(\mathfrak{P}, \mathbf{a})$ is an algebraic Hecke character $\chi_{\mathbf{a}}$ of k with values in k . Let $\langle \mathbf{a} \rangle$ be the representative between 0 and 1 of $\mathbf{a}$ in $\mathbb{Z}/N$. If $\mathbf{a} = \sum n(a)\delta_a$ satisfies

$$(*) \text{ For every } u \in (\mathbb{Z}/N)^*, \sum n(a)\langle ua \rangle = 0,$$

Weil's determination of the algebraic part of $\chi_{\mathbf{a}}$ implies that $\chi_{\mathbf{a}}$ has finite order; write $\chi_{\mathbf{a}}$ also for the character of $\text{Gal}(\tilde{\mathbb{Q}}/k)$ whose value at geometric Frobenius at $\mathfrak{P}$ is $\chi_{\mathbf{a}}(\mathfrak{P})$.

Put $\Gamma(\mathbf{a}) = \prod_a \Gamma(\langle \mathbf{a} \rangle)^{n(a)}$. A first part of the conjecture, the algebraicity of $\Gamma(\mathbf{a})$ when $\mathbf{a}$ satisfies $(*)$, was proved by Koblitz and Ogus; see the appendix. More precisely, one would like:

Conjecture 8.11. If $\mathbf{a}$ satisfies $(*)$, then $\sigma\Gamma(\mathbf{a}) = \chi_{\mathbf{a}}(\sigma) \cdot \Gamma(\mathbf{a})$ for every $\sigma \in \text{Gal}(\tilde{\mathbb{Q}}/k)$.

If $\mathbf{a}$ is invariant under a subgroup H of $(\mathbb{Z}/N)^*$, 8.11 can be sharpened by replacing $\text{Gal}(\tilde{\mathbb{Q}}/k)$ with $\text{Gal}(\tilde{\mathbb{Q}}/k^H)$, and by using SGA 4^{1/2} 6.5 to define a character of this group with values in the roots of unity of k^H . This is done below.

8.12. Let H be a subgroup of $(\mathbb{Z}/N)^*$, $\mathfrak{P}$ a prime ideal of k^H prime to N , κ its residue field, and $\Psi: \kappa \rightarrow \mathbb{C}^*$ a nontrivial additive character. For $a \in N^{-1}\mathbb{Z}/\mathbb{Z} - \{0\}$, let $\mathfrak{P}_{a,i}$ be the prime ideals of $k^H(\exp(2\pi ia))$ above $\mathfrak{P}$. If κ' is the residue field at $\mathfrak{P}_{a,i}$ and $|\kappa'| = q'$, write $g(\mathfrak{P}_{a,i}, a, \Psi)$ for the Gauss sum

$$- \sum_{x \in \kappa'^*} t(x^{-a(q'-1)}) \Psi(\text{Tr}_{\kappa'/\kappa} x).$$

The product $g(\mathfrak{P}, \mathbf{a}, \Psi) = \prod_i g(\mathfrak{P}_{a,i}, a, \Psi)$ depends only on the H -orbit of $\mathbf{a}$.

Let $\mathbf{a} = \sum n(a)\delta_a$ be H -invariant and satisfy $\sum n(a)a = 0$. For every H -orbit O in $N^{-1}\mathbb{Z}/\mathbb{Z} - \{0\}$, write $n(O)$ for the constant value of $n(a)$ on O , and put

$$g(\mathfrak{P}, \mathbf{a}) = \prod_{a \bmod H} g(\mathfrak{P}, a, \Psi)^{n(a)}.$$

As a function of $\mathfrak{P}$, $g(\mathfrak{P}, \mathbf{a})$ is an algebraic Hecke character $\chi_{\mathbf{a}}$ of k^H with values in k^H . To prove this, by additivity one reduces to $n(a) \geq 0$. Apply [3, 6.5] with $F = k^H$, $\tilde{F} = \tilde{\mathbb{Q}}$, k equal to the present k , and I a disjoint union of H -orbits, with $n(O)$ copies of O . For $i \in I$ whose image in $N^{-1}\mathbb{Z}/\mathbb{Z}$ is a/N , take λ_i to be the composite

$$\hat{\mathbb{Z}}(1)_F \rightarrow (\mathbb{Z}/N)(1)_F = \mu_N(k) \rightarrow^{x \mapsto x^a} k^*.$$

The resulting Hecke character is the product of $\chi_{\mathbf{a}}$ with the "signature" character of the

permutation representation of H on I'' . A special case of this result already appears in Weil [17]. The character χ_a of 8.10 is the composite of the χ_a above with the norm N_{k/k^H} .

Conjecture 8.13. Let $\mathfrak{P}$ be a prime ideal of k^H prime to N , and let $F_{\mathfrak{P}}$ be a geometric Frobenius at $\mathfrak{P}$. If the H -invariant a satisfies (*), then

$$F_{\mathfrak{P}}\Gamma(a) = g(\mathfrak{P}, a) \cdot \Gamma(a).$$

In particular, if the group of roots of unity of k^H has order N' , then $\Gamma(a)^{N'} \in k^H$.

One can deduce from 8.13 the following, apparently more general, variant. Replace condition (*) by

$$(*)' \sum n(a)\langle ua \rangle = k, \text{ an integer independent of } u \in (\mathbb{Z}/N)^*.$$

The Hecke character $N\mathfrak{P}^{-k} \cdot g(\mathfrak{P}, a)$ then has finite order. Identify it with a character χ of $\text{Gal}(\bar{\mathbb{Q}}/k^H)$; one expects $\sigma((2\pi i)^{-k} \Gamma(a)) = \chi(\sigma) \cdot ((2\pi i)^{-k} \Gamma(a))$.

8.14. If k is a totally imaginary quadratic extension of a totally real field—that is, in the terminology used here, a field of CM type—Shimura [13] determined the critical values of the L -functions of algebraic Hecke characters of k up to multiplication by an algebraic number. He expresses them in terms of periods of CM abelian varieties with complex multiplication by k . In the remainder of the section, we show that his theorem is compatible with Conjecture 2.8, which expresses them in terms of periods of rank-one motives over k .

8.15. Let χ and $M(\chi)$ be as in 8.1. Exclude the case 8.5 and suppose $R_{k/\mathbb{Q}}M(\chi)$ satisfies 1.7. Then k is totally imaginary, and the Hodge numbers $h^{p,p}$ vanish: with the notation of 8.1 and 8.2, no n_i equals $w/2$. Our first task is to compute $c^+ R_{k/\mathbb{Q}}M(\chi)$ in terms of the periods $p'(\chi; \sigma, \tau)$. Recall that it is the determinant of the isomorphism of $E \otimes \mathbb{C}$ -modules

$$I^+ : H_B^+ R_{k/\mathbb{Q}}M(\chi) \otimes \mathbb{C} \xrightarrow{\sim} H_{\text{DR}}^+ R_{k/\mathbb{Q}}M(\chi) \otimes \mathbb{C},$$

calculated in bases defined over E .

Choose the e_{σ} of 8.6 so that $F_{\infty}e_{\sigma} = e_{c\sigma}$. The vectors $e_{\sigma} \pm e_{c\sigma}$ then form a basis of $H_B^+ R_{k/\mathbb{Q}}M(\chi) \subset H_{\text{BR}k/\mathbb{Q}}M_{\chi} = \bigoplus_{\sigma \in S} H_{\sigma}M(\chi)$. Note that $d^+ = d^- = \frac{1}{2}[k:\mathbb{Q}]$. Let $\bar{S}$ be the quotient of S by $\text{Gal}(\mathbb{C}/\mathbb{R})$. To calculate $c^+ = \det(I^+)$, use the basis $(e_{\sigma} + e_{c\sigma})$ of H_B^+ , indexed by $\bar{S}$.

By 8.1(iv), the Hodge filtration of $H_{\text{DR}k/\mathbb{Q}}M(\chi) = H_{\text{DR}M}(\chi)$ is read from its $k \otimes E$ -module structure. Let $(k \otimes E)^+$ be the direct factor of $k \otimes E$ that is the product of the K_i with $n_i < w/2$. The quotient $H_{\text{DR}}^+ R_{k/\mathbb{Q}}M(\chi)$ is the corresponding direct factor:

$$H_{\text{DR}}^+ R_{k/\mathbb{Q}}M(\chi) = H_{\text{DR}M}(\chi) \otimes_{k \otimes E} (k \otimes E)^+.$$

Let ω be as in 8.7, and use the $k \otimes E \otimes \mathbb{C} = \mathbb{C}^{S \times J}$ -module structure of $H_{\text{DR}M}(\chi)$ to decompose ω as $\omega = \sum \omega_{\sigma, \tau}$. By definition $I(e_{\sigma}) = \sum_{\tau} p'(\sigma, \tau)^{-1} \omega_{\sigma, \tau}$, hence

$$I^+(e_{\sigma} + e_{c\sigma}) = \sum_{n(\sigma, \tau) < w/2} p'(\sigma, \tau)^{-1} \omega_{\sigma, \tau} + \sum_{n(c\sigma, \tau) < w/2} p'(c\sigma, \tau)^{-1} \omega_{c\sigma, \tau}.$$

This is the sum, indexed by $\tau \in J$, of terms $p'(\sigma, \tau)^{-1} \omega_{\sigma, \tau}$ when $n(\sigma, \tau) < w/2$, and $p'(c\sigma, \tau)^{-1} \omega_{c\sigma, \tau}$ when $n(\sigma, \tau) > w/2$. For $\bar{\sigma} \in \bar{S}$ represented by σ , put

$$\omega_{\bar{\sigma}} = \sum_{n(\sigma, \tau) < w/2} \omega_{\sigma, \tau} + \sum_{n(c\sigma, \tau) < w/2} \omega_{c\sigma, \tau}.$$

The vectors $I^+(e_\sigma + e_{c\sigma})$ are multiples of the $\omega_{\bar{\sigma}}$ by elements of $E \otimes \mathbb{C}$; thus the $\omega_{\bar{\sigma}}$ form a basis of H_{DR}^+ . In the bases $e_\sigma + e_{c\sigma}$ and $\omega_{\bar{\sigma}}$, the matrix of I^+ is diagonal, and its determinant $\det'(I^+) \in (E \otimes \mathbb{C})^* = \mathbb{C}^{*J}$ has coordinates

$$\det'(I^+)_\tau = \prod_{n(\sigma, \tau) < w/2} p'(\sigma, \tau)^{-1}.$$

Consider the maps $E^{\bar{S}} \rightarrow E^{\bar{S}}$ given by $1_{\bar{\sigma}} \mapsto 1_\sigma + 1_{c\sigma}$, for $\bar{\sigma}$ the image of σ ; the isomorphism $E^{\bar{S}} \otimes \mathbb{C} \xrightarrow{\sim} k \otimes E \otimes \mathbb{C}$ induced by $k \otimes \mathbb{C} \cong \mathbb{C}^{\bar{S}}$; and the projection of $k \otimes E$ onto $(k \otimes E)^+$. Their composite is an isomorphism of $E \otimes \mathbb{C}$ -modules

$$E^{\bar{S}} \otimes \mathbb{C} \xrightarrow{\sim} (k \otimes E)^+ \otimes \mathbb{C}.$$

Let $D(\chi)$ be its determinant, calculated in bases defined over E on both sides. Identifying H_{DR}^+ with $k \otimes E$ by means of the basis ω , this is the determinant of the identity map on $H_{\text{DR}}^+ \otimes \mathbb{C}$, calculated in the basis $\omega_{\bar{\sigma}}$ at the source and in a basis defined over E at the target. Let $D_\tau(\chi)$ denote its components in $\mathbb{C}^J$. Since $c^+ = \det'(I^+) \cdot D(\chi)$, one obtains:

PROPOSITION 8.16. One has

$$c^+ R_{k/\mathbb{Q}} M(\chi) = \left(\prod_{n(\sigma, \tau) < w/2} p'(\sigma, \tau)^{-1} \cdot D_\tau(\chi) \right)_{\tau \in J}.$$

REMARK 8.17. Suppose k is of CM type, a quadratic extension of the totally real field k_0 . Then $\bar{S}$ identifies with the set of complex embeddings of k_0 , and the diagram

$$\begin{array}{ccccc} E^{\wedge\{\bar{S}\}} \otimes \mathbb{C} & \longrightarrow & k_0 \otimes E \otimes \mathbb{C} & & \\ \downarrow & & \downarrow & & \\ E^{\wedge S} \otimes \mathbb{C} & \longrightarrow & k \otimes E \otimes \mathbb{C} & \longrightarrow & (k \otimes E)^+ \otimes \mathbb{C} \end{array}$$

commutes. The composite $k_0 \otimes E \rightarrow (k \otimes E)^+$ is therefore an isomorphism, and $D(\chi)$ is also the determinant of $E^{\bar{S}} \otimes \mathbb{C} \xrightarrow{\sim} k_0 \otimes E \otimes \mathbb{C}$ obtained by extending scalars from $\mathbb{Q}$ to E in the isomorphism $\mathbb{C}^{\bar{S}} \xrightarrow{\sim} k_0 \otimes \mathbb{C}$. Thus $D(\chi)$, well defined modulo E^* , has a representative in $(\mathbb{Q} \otimes \mathbb{C})^* = \mathbb{C}^* \subset (E \otimes \mathbb{C})^*$, namely the determinant of the inverse of the matrix $(\sigma\alpha)$, where $\sigma \in \bar{S}$ and α runs through a basis of k_0 over $\mathbb{Q}$. The isomorphism $\mathbb{C}^{\bar{S}} \xrightarrow{\sim} k_0 \otimes \mathbb{C}$ transforms the quadratic form $\sum x_\sigma^2$ into $\text{Tr}(xy)$. Hence $(\det(\sigma\alpha))^2$ identifies with the discriminant of k_0 :

$D(\chi) \sim$ a square root of the discriminant of k_0 .

8.18. Let $p''(\chi; \sigma, \tau)$ denote the image of $p'(\chi; \sigma, \tau)$ in $\mathbb{C}^*/\bar{\mathbb{Q}}^*$. It depends only on χ , σ , and τ . If an algebraic homomorphism $\eta: k^* \rightarrow E^*$ satisfies (8.2.1), some power of it is the algebraic part of a Hecke character: $\eta^N = \chi_{\text{alg}}$. Moreover, if $\chi'_{\text{alg}} = \chi''_{\text{alg}}$, then χ' and χ'' differ only by a finite-order character, and $\chi'^M = \chi''^M$ for a suitable M . It follows from (8.1.1) that $p''(\chi^M; \sigma, \tau) = p''(\chi; \sigma, \tau)^M$, which permits the following unambiguous definition:

$p(\eta; \sigma, \tau) = p(\chi; \sigma, \tau)^{1/N}$, when $\eta^N = \chi_{\text{alg}}$.

These periods satisfy the following formalism:

$$(8.18.1) \quad p(\eta' \eta''; \sigma, \tau) = p(\eta'; \sigma, \tau) p(\eta''; \sigma, \tau).$$

(8.18.2) $p(\eta; \sigma, \tau)$ is unchanged when E is replaced by an extension E' and τ by one of its extensions to E' .

(8.18.3) $p(\eta; \sigma, \tau)$ is unchanged when k is replaced by an extension k' , σ by one of its extensions to k' , and χ by $\chi \circ N_{k'/k}$.

(8.18.4) If α is an automorphism of k and β an automorphism of E , then

$$p(\eta; \sigma, \tau) = p(\beta \eta \alpha^{-1}; \sigma \alpha^{-1}, \tau \beta^{-1}).$$

(8.18.5) The complex conjugate of $p(\eta; \sigma, \tau)$ is $p(\eta; \bar{\sigma}, \bar{\tau})$.

(8.18.6) For $k = F = \mathbb{Q}$, $p(\text{Id}, \text{Id}, \text{Id}) = 2\pi i$.

Formulas (8.15.1)–(8.15.3) follow from (8.1.1)–(8.1.3); (8.18.4) and (8.18.5) follow by transport of structure; and (8.18.6) follows from (8.5).

8.19. Let $\eta: k^* \rightarrow E^*$ be a homomorphism satisfying (8.2.1). Suppose also that $n(\eta; \sigma, \tau)$ is never $w/2$, which makes it possible to define $(k \otimes E)^+$ as in 8.15. Define $\eta^*: E^* \rightarrow k^*$ by

$$\eta^*(y) = \det_k(1 \otimes y, (k \otimes E)^+).$$

This homomorphism again satisfies (8.2.1), and

$$n(\eta^*; \sigma, \tau) = 1 \text{ if } n(\eta; \sigma, \tau) < w/2, 0 \text{ if } n(\eta; \sigma, \tau) > w/2.$$

If η^* is the algebraic part of a Hecke character χ^* , then $M(\chi^*)$ is the H^1 of an abelian variety over E with complex multiplication by k , whose Lie algebra, as a $k \otimes E$ -module, is isomorphic to $(k \otimes E)^+$.

PROPOSITION 8.20. Under the hypotheses and notation of 8.19, regard E as a subfield of $\mathbb{C}$ and write 1 for the identity embedding $E \hookrightarrow \mathbb{C}$. Then

$$\prod_{n(\sigma, 1) < w/2} p(\eta; \sigma, 1) = \prod_{\sigma} p(\eta^*; 1, \sigma)^{n(\eta; \sigma, 1)}.$$

If $\iota: E \rightarrow E' \subset \mathbb{C}$ is a finite extension of E , the left-hand side does not change when η is replaced by $\iota\eta$, by (8.18.2). One has $(\iota\eta)^* = \eta^* \circ N_{E'/E}$; by (8.18.3), the right-hand side is likewise unchanged.

If $\iota: k \rightarrow k'$ is a finite extension of degree d , the left-hand side is raised to the d -th power when k is replaced by k' and η by $\eta \circ N_{k'/k}$: a complex embedding of k is induced by d embeddings of k' , and (8.18.3) applies. The same holds for the right-hand side, by (8.18.2) and the equality $(\eta \circ N_{k'/k})^* = \iota\eta^*$.

These compatibilities reduce us to the case in which E is Galois and k is isomorphic to E . For each isomorphism ω from k to E , put $n(\omega) = n(\eta; 1 \circ \omega, 1)$. Writing the group of homomorphisms $k^* \rightarrow E^*$ additively, $\eta = \sum n(\omega)\omega$. Since $p(\eta; 1 \circ \omega, 1) = p(\eta \circ \omega^{-1}; 1, 1)$, by (8.15.4), one obtains

$$(8.20.1) \quad \prod_{n(\sigma, 1) < w/2} p(\eta; \sigma, 1) = \prod_{n(\omega) < w/2} p(\eta \circ \omega^{-1}; 1, 1) = p(\sum_{n(\omega) < w/2} \eta \circ \omega^{-1}; 1, 1).$$

On the other hand,

$$\sum_{n(\omega) < w/2} \eta \circ \omega^{-1} = \sum_{n(\omega_1) < w/2; \omega_2} n(\omega_2) \omega_2 \circ \omega_1^{-1} = \sum_{\omega_2} n(\omega_2) \sum_{n(\omega_1) < w/2} \omega_2 \circ \omega_1^{-1} = \sum_{\omega_2} n(\omega_2) \omega_2 \circ \eta^*.$$

This permits (8.20.1) to be continued as

$$= \prod_{\omega} p(\omega \circ \eta^*; 1, 1)^{n(\omega)} = \prod_{\omega} p(\eta^*; 1, 1 \circ \omega)^{n(\omega)} = \prod_{\sigma} p(\eta^*; 1, \sigma)^{n(\eta; \sigma, 1)},$$

(a new application of 8.15.4), and proves 8.20.

8.21. Combining 8.16 and 8.20, one obtains, for the component of index 1 of

$$c^+ R_{k/Q} M(\chi) \in (E \otimes C)^*/E^* = C^{*J}/E^*,$$

the following expression, modulo $\bar{Q}^*$:

$$c_1^+ R_{k/Q} M(\chi) \sim \prod_{\sigma} p(\chi^*_{\text{alg}}; 1, \sigma)^{-n(\chi; \sigma, 1)}.$$

If χ (i.e., $M(\chi)$) is critical, Conjecture 2.8 therefore asserts that

$$L(1 \circ \chi, 0) \sim \prod_{\sigma} p(\chi^*_{\text{alg}}; 1, \sigma)^{-n(\chi; \sigma, 1)} \pmod{\bar{Q}^*}.$$

For E sufficiently large, χ^*_{alg} is the algebraic part of a Hecke character χ^* , and the periods are interpreted as periods of abelian integrals (8.19), (8.3). The resulting statement is the one Shimura proved when k is of CM type, or is abelian over a field of CM type (with a restriction on the weight).

REMARK 8.22. If $\eta : k^* \rightarrow E^*$ satisfies (8.2.1), there are subfields k' of k and E' of E , each either of CM type or equal to $\mathbb{Q}$, an inclusion $\iota : E' \rightarrow E$, and a factorization $\eta = \eta' N_{k/k'}$. One then has $p(\eta; \sigma, \tau) = p(\eta'; \sigma | k', \tau | E')$.

If now k and E are of CM type (or $\mathbb{Q}$), and c again denotes their complex conjugation, then $c\sigma = \sigma c$, $c\tau = \tau c$, and $\eta c = c\eta$; hence

$$p(\eta; \sigma, \tau) = p(\eta; c\sigma, c\tau) = p(\eta; \sigma c, \tau c) = p(c\eta c^{-1}; \sigma, \tau) = p(\eta; \sigma, \tau).$$

Thus the periods, a priori in $C^*/\bar{Q}^*$, are real, i.e., lie in $R^*/(\bar{Q}^* \cap R^*)$.

REMARK 8.23. Let G be the Galois group of the union of the CM-type extensions of $\mathbb{Q}$ in $\bar{\mathbb{Q}} \subset \mathbb{C}$, and let $c \in G$ be complex conjugation. It is a central element of G . If φ is a locally constant integer-valued function on G , put $\varphi^*(x) = \varphi(xc)$. Suppose that $\varphi + \varphi^*$ is constant. Let G_1 be a finite quotient of G through which φ factors as a function φ_1 on G_1 , and let k be the corresponding field. The hypothesis means that the endomorphism $\sum \varphi_1(\sigma) \sigma$ of k^* satisfies (8.2.1). The period $p(\sum \varphi_1(\sigma) \sigma; 1, 1)$ is independent of the choice of G_1 ; put $P(\varphi) = p(\sum \varphi_1(\sigma) \sigma; 1, 1)$. The functional P is a homomorphism into $C^*/\bar{Q}^*$ from the group of locally constant integer-valued functions on G satisfying $\varphi + \varphi^* = \text{constant}$.

Appendix by N. Koblitz and A. Ogus. Algebraicity of some products of values of the Γ function. Let $A_N = N^{-1}\mathbb{Z}/\mathbb{Z} - \{0\}$, and let $U_N = (\mathbb{Z}/N\mathbb{Z})^*$ operate on A_N in the obvious way. If $f : A_N \rightarrow \mathbb{C}$, define $\langle f \rangle : U_N \rightarrow \mathbb{C}$ by

$$\langle f \rangle(u) = \sum_{a \in A_N} \langle a \rangle f(ua),$$

where $\langle a \rangle$ denotes the representative of a between 0 and 1.

We first compute $H_Q = \{f : A_N \rightarrow \mathbb{Q} : \langle f \rangle \text{ is constant}\}$.

EXAMPLE. If n is a divisor of N and $a \in A_N$, consider the set

$$S_{n,a} = [\{a + k/n : k = 0, 1, \dots, n-1\} \cup \{-na\}] \cap A_N.$$

It is easy to see that its characteristic function $\varepsilon_{n,a}$ is in H_Q . Note that $S_{1,a} = \{a, -a\}$.

PROPOSITION. H_Q is generated by $\{\varepsilon_{n,a} : n = 1 \text{ or } n \text{ is prime and } na \neq 0\}$.

PROOF. The orbits of A_N under the action of U_N correspond to the divisors d of N with $1 < d \leq N$: each orbit can be written uniquely in the form $U_N(1/d)$. The stabilizer subgroup I_d of $1/d$ is $\{u \in U_N : u \equiv 1 \pmod{d}\}$ and the orbit of $1/d$ is canonically isomorphic to U_d . Thus, a function f on A_N is determined by the collection of functions $f_d : U_d \rightarrow \mathbb{Q}$ defined by $f_d(v) = f(v/d)$.

To prove the proposition, we first complexify, so as to be able to work with characters of U_N .

LEMMA. If $f : A_N \rightarrow \mathbb{C}$ and if χ is a character of U_N , then the inner product of $\langle f \rangle$ and χ is given by:

$$\langle \langle f \rangle | \chi \rangle = -\sum_d L(0, \chi_d) |I_d| \cdot \langle f_d | \chi_d \rangle,$$

where the sum is taken over those divisors d of N such that χ is pulled back from a character χ_d of U_d .

PROOF. We have

$$\langle \langle f \rangle | \chi \rangle = \sum_{u \in U_N} \langle f \rangle(u) \bar{\chi}(u) = \sum_{a \in A_N} \sum_{u \in U_N} \langle a \rangle f(ua) \bar{\chi}(u) = \sum_{d|N} \sum_{v \in U_d} \sum_{u \in U_N} \langle v/d \rangle f_d(u_{dv}) \bar{\chi}(u),$$

where $u_d \in U_d$ is the image of u .

Now if we choose a set $U'_d \subseteq U_N$ of coset representatives for $U_N \rightarrow U_d$, we can write:

$$\sum_{u \in U_N} f_d(u_{dv}) \bar{\chi}(u) = \sum_{u' \in U'_d} \sum_{u \in I_d} f_d(u'_{dv}) \bar{\chi}(u') \bar{\chi}(u) = \sum_{u' \in U'_d} f_d(u'_{dv}) \bar{\chi}(u') \sum_{u \in I_d} \bar{\chi}(u).$$

Of course, this sum is zero unless χ is trivial on I_d , i.e., unless χ is pulled back from a character χ_d of U_d , in which case it is

$$\sum_{u' \in U'_d} f_d(u'_{dv}) \bar{\chi}_d(u') |I_d| = \sum_{w \in U_d} f_d(wv) \bar{\chi}_d(w) \chi_d(v) |I_d| = \chi_d(v) |I_d| \langle f_d | \chi_d \rangle.$$

If we substitute this into the above expression for $\langle \langle f \rangle | \chi \rangle$ and note that $\langle v/d \rangle = |v|/d$, where $|v|$ is the smallest positive representative of $v \in U_d$, we find:

$$\langle \langle f \rangle | \chi \rangle = \sum_d \sum_{v \in U_d} |v| \chi_d(v) |I_d| d^{-1} \langle f_d | \chi_d \rangle = -\sum_d L(0, \chi_d) |I_d| \langle f_d | \chi_d \rangle,$$

as claimed. $\square$

To prove the proposition, we let d_f be the largest divisor of N such that $f_{d_f} \neq 0$; it suffices to prove that any f in H_Q can be written as $g + f'$, with g a linear combination of the ε 's and $d_{f'} < d_f$. Let $d = d_f$; we shall show below that f_d is a linear combination of functions h_1 which factor through $U_d \rightarrow U_d/\{\pm 1\}$, and functions h_p which factor through $U_d \rightarrow U_d/p$ for some prime divisor p of d , $p \neq d$. Any function of the first type is invariant under ± 1 , and hence the corresponding function on the orbit $U_N(1/d)$ is a linear combination of $\varepsilon_{1,a}$'s. A function of the second type is invariant under the kernel K of $U_d \rightarrow U_d/p$, which has order p if p divides d/p and order $p - 1$ otherwise. In the first case, $K = \{1 + kd/p : k = 0, \dots, p - 1\}$, and in

the second, it is this same set with one element deleted, namely, the value of $1 + kd/p$ which is divisible by p . Thus, the set $S_{p,w/d} = (Kw)/d$, with the addition of one or two elements in the orbit $U_N(p/d)$. Hence it is clear that the corresponding function on the orbit $U_N(1/d)$ can be written as a linear combination of $\varepsilon_{p,a}$'s and functions supported on the orbit of p/d , and we have obtained our desired decomposition $f = g + f'$.

It remains for us to prove the claim about f_d . If d is twice an odd number the map $U_d \rightarrow U_{d/2}$ is an isomorphism, and the claim is trivial. In the other cases, choose a primitive odd character χ_d of U_d , and let χ be the pull-back of χ_d to U_N . Since χ is nontrivial and $\langle f \rangle$ is constant, $\langle f \rangle \mid \chi = 0$. Let us look at the expression for this inner product provided by the lemma; the only nonzero terms come from those f_e such that $f_e \neq 0$ and $I_e \subseteq \text{Ker}(\chi)$. But then $\text{Ker}(\chi)$ contains I_e and I_d , hence also $I_{d|e} = I_m$, where $m = \text{g.c.d.}(d, e)$. Since χ is primitive, $m = d$ and d divides e , and since $f_e = 0$ for $e > d$,

$$0 = \langle f \rangle \mid \chi = -L(0, \chi_d) |I_d| (f_d \mid \chi_d).$$

Since χ_d is odd and primitive, we conclude that $(f_d \mid \chi_d) = 0$, i.e., f_d is orthogonal to every odd primitive character of U_d . It follows that f_d can be written as a linear combination of characters which are even (factor through $U_d/\{\pm 1\}$) or imprimitive (factor through some $U_{d/p}$). When $d = p$, we should also remark that the constant functions on U_p are also invariant under ± 1 , and hence are already covered by the first case. $\square$

THEOREM. Suppose that $f : A_N \rightarrow \mathbb{Q}$ is such that $\langle f \rangle$ is constant: $\sum_a \langle a \rangle f(ua) = k$ for all $u \in U_N$. Then

$$\Gamma(f) := \pi^{-k} \prod_a \Gamma(\langle a \rangle)^{f(a)}$$

is an algebraic number.

PROOF. The map $\langle \rangle : H_Q \rightarrow \mathbb{Q}$ sending any f to $\sum_a \langle a \rangle f(a)$ is evidently linear, and hence $\Gamma(f + g) = \Gamma(f)\Gamma(g)$, for f and g in H_Q . By the proposition, any f in H_Q is a $\mathbb{Q}$ -linear combination of the ε 's; hence nf is a sum of ε 's for some n . Since $\Gamma(f)^n = \Gamma(nf)$, we are reduced to checking the theorem for the ε 's described in the proposition. Noting that $\langle \varepsilon_{1,a} \rangle = 1$ and that $\langle \varepsilon_{p,a} \rangle = (p + 1)/2$ if $pa \neq 0$, one has only to appeal to the identities:

$$\Gamma(x)\Gamma(1-x) = \pi(\sin \pi x)^{-1},$$

$$\sum_{k=1}^{p-1} \Gamma(x + k/p) = p^{1/2-x} (2\pi)^{(p-1)/2} \Gamma(px). \quad \square$$

REMARK. Kubert has recently obtained a much more precise result expressed in a different terminology from which it follows that, if H_Q is $\mathbb{Z}$ -valued, then $2f$ is a $\mathbb{Z}$ -linear combination of the ε 's (to appear).

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INSTITUT DES HAUTES ÉTUDES SCIENTIFIQUES, BURES SUR YVETTE, FRANCE